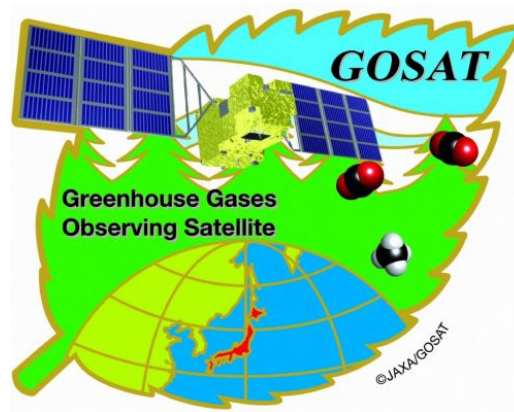




GOSAT Reverse Engineering Study

Final Report

Team 30

P. Code	Surname	Name
11119843	Challita	Jack
11179425	Macchia	Bruno
11153986	Mazurek	Krzysztof
10901276	Ponti	Alessandro
11161421	Thomas	Bastien
11141405	Tomczykowski	Tymon

Space System Engineering and Operations

Prof: M. Lavagna

L. Bianchi, M. Bussolino

M.Sc Space Engineering

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List of Acronyms

AKE Attitude Knowledge Error.

AOCE Attitude and Orbit Control Electronics.

AOCS Attitude and Orbit Control System.

APE Attitude Performance Error.

BOL Beginning Of Life.

CAI Cloud and Aerosol Imager.

CDMS Command and Data Management System.

CFRP Carbon Fiber Reinforced Polymer.

ConOp Conceptual Operation.

DET Direct Energy Transfer.

DoD Depth of Discharge.

EOC Earth Observation Center.

EOL End of Life.

EPS Electrical Power System.

ES Earth Sensor.

FDIR Fault Detection, Isolation, and Recovery.

FDS Flight Dynamics System.

FSS Fine Sun Sensor.

FTS Fourier Transform Spectrometer.

GOSAT Greenhouse gases Observing SATellite.

GPSR GPS Receiver.

HDRM Hold Down & Release Mechanism.

HIT Heavy Ion Telescope.

HKDR Housekeeping Data Recorder.

IF Interface.

IR Infrared.

IRU Inertial Reference Unit.

ITU International Telecommunication Union.

JAXA Japan Aerospace Exploration Agency.

LEO Low Earth Orbit.

LPT Light Particle Telescope.

LTDN Local Time of Descending Node.

MDC Mission Duty Cycle.

MDHS Mission Data Handling Subsystem.

MDP Mission Data Processor.

MLI Multi Layer Insulation.

MTQ Magnetic Torquer.

NIES National Institute for Environmental Studies.

OBC On-Board Computer.

OBDH On Board Data Handling.

OSR Optical Solar Reflectors.

PL Payload.

PS Propulsion System.

RCS Reaction Control System.

RIM Remote Interface Module.

RTOS Real-Time Operating System.

RW Reaction Wheel.

S/C Spacecraft.

SAA Sun Aspect Angle.

SCS Satellite Control System.

SSO Sun Synchronous Orbit.

SSPA Solid State Power Amplifier.

STT Star Tracker.

SvalSat Svalbard Satellite Station.

SWIR Short Wavelength InfraRed.

TANSO Thermal And Near infrared Sensor for carbon Observation.

TBD To Be Decided.

TCS Thermal Control System.

TEDA Technical Data Acquisition Equipment.

TIR Thermal InfraRed.

TMTC Tracking Telemetry and Telecommand.

TT&C Telemetry, Tracking and Command.

UV Ultraviolet.

1 GOSAT Mission

1.1 Introduction

The Greenhouse gases Observing SATellite (GOSAT) represents a fundamental milestone in global environmental monitoring from space. Launched on January 23, 2009, as a joint initiative by Japan Aerospace Exploration Agency (JAXA), the Ministry of the Environment, and National Institute for Environmental Studies (NIES), the mission was designed to track greenhouse gases on a global scale. With the data collected, scientists can accurately estimate carbon dioxide (CO_2) and methane (CH_4) sources and sinks on a sub-continental scale, directly supporting the Kyoto Protocol. To achieve these objectives, the spacecraft utilizes two primary instruments: the TANSO-FTS spectrometer and the TANSO-CAI radiometer. Operating well beyond its five-year nominal lifespan, GOSAT continues to deliver highly accurate, well-calibrated spectral data, laying the groundwork for future climate observation missions.



Figure 1: Gosat "Ibuki" [1]

1.2 Mission High Level Goals

The GOSAT mission had two main goals, one scientific and one technical, which are described in Table 1.

Table 1: GOSAT Mission High Level Goals [2]

Goals	Reasons
Map the global distribution of Greenhouse Gases net emissions to identify sources and sinks at subcontinental level.	By addressing gaps in the ground-station network the GOSAT Project will provide the reliable, objective data needed to verify international compliance with the Kyoto Protocol and subsequent climate agreements.
Contribute in the advancement of earth observation technologies for future missions.	GOSAT will act as a technological pioneer, developing and validating innovative geosensing instruments, while also proving the reliability of key components.

1.3 Mission Drivers

The mission drivers are highly constraining requirements that significantly influence the design choices made at any phase of the project. The driver for GOSAT mission is:

Spacecraft Reliability

During the previous JAXA Earth observation missions ADEOS-I and ADEOS-II, failures of the solar arrays and power system occurred, leading to the premature end of the missions. GOSAT is the first satellite developed after these setbacks. For this reason, achieving high reliability is the main priority of the design process [2]. The following recommendations should be taken into account at every stage of the project:

- Maximum utilization of flight-proven components.
- Complete elimination of single-point failures.
- Strategic adoption of functional redundancy.
- Thorough Mission Duty Cycle (MDC) tests.

1.4 Mission Functional Analysis

The mission objectives are achieved through a set of defined tasks, which can be organized and visualized using the functional tree in Figure 2.

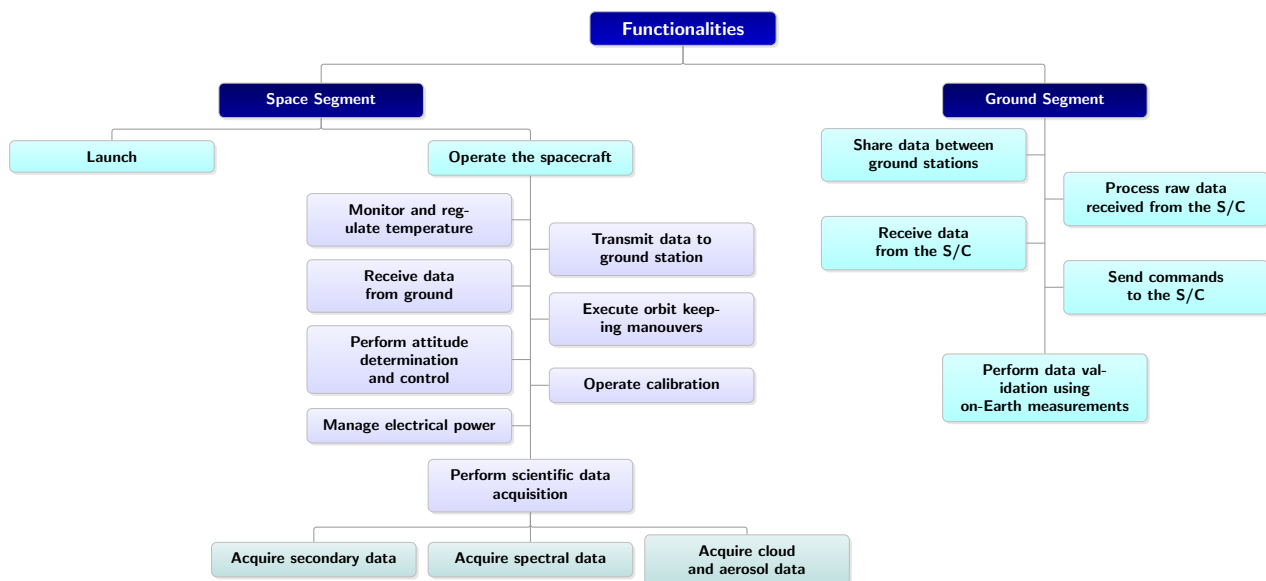


Figure 2: Functional Tree [3, 4]

1.5 Mission & System Level Requirements

The mission and system level requirements for the GOSAT mission are derived from the high-level objectives and mission drivers. These requirements represent a quantitative translation of the mission objectives into technical constraints that guide the design of both the space and ground segments.

1. Mission Requirements

- The system shall measure the global distribution of atmospheric column concentrations of CO_2 and CH_4 with an accuracy of [1]:
 - $\leq 1\%$ for CO_2 (approximately 4 ppm over a 3-month average)

- $\leq 2\%$ for CH_4 (approximately 34 ppb).
- The system shall achieve a spatial resolution in the range of 100–1000 km.
- The mission shall operate for a nominal duration of at least 5 years.

2. Orbit and Coverage Requirements

- The spacecraft shall operate in a Sun-synchronous orbit to ensure consistent illumination conditions for spectroscopic measurements.
- The orbit shall provide global coverage with a 3-day repeat cycle and with a recurrence error of less than ± 2.5 km [1].

3. Attitude Determination and Control Requirements

- The system shall provide pointing accuracy To Be Decided (TBD) and stability to support nadir observations of the Earth and controlled scanning required by the payload instruments.
- The system shall be able to point specific targets required for instruments calibration.

4. Environmental Requirements

- The spacecraft shall maintain all subsystems and payload components within their operational temperature ranges.
- The spacecraft shall withstand the radiation environment of Low Earth Orbit (LEO), including solar radiation and cosmic particles.

5. Data Handling and Communication Requirements

- The spacecraft shall include sufficient onboard memory of 48 GB to store data between ground station contacts [1].
- The system shall enable reliable Telemetry, Tracking and Command (TT&C) operations.
- The system shall support periodic downlink of scientific data with a bandwidth of 120 Mbit/s [1].

6. Power System Requirements

- The spacecraft shall generate sufficient electrical power to support payload and platform operations.
- The system shall include sufficient energy storage capabilities to ensure operation during eclipse periods.

7. Orbit Control Requirements

- The spacecraft shall be capable of maintaining its orbital parameters and counteracting environmental disturbances.

8. Launch and Structural Requirements

- The spacecraft shall be compatible with the H-IIA launch vehicle interface and constraints.
- The system shall withstand launch loads including accelerations, vibrations, and acoustic environments.

9. End-of-life

- The system shall support passivation and end-of-life disposal or deorbiting strategy in compliance with space debris mitigation guidelines.

1.6 Mission Phases and Correlation with ConOps

The GOSAT mission spans over five years, divided into six phases, each aligned with the Conceptual Operations (ConOps) as detailed in [1] and [3].

1. Launch

Mitsubishi Heavy Industries, Ltd. and the JAXA will launch the GOSAT aboard the H-IIA Launch Vehicle from the Tanegashima Space Center. The launch phase will last approximately 16 minutes, after which the spacecraft will be inserted into a sun-synchronous orbit.

2. Critical Phase

The critical operation phase is defined as the period of preparation for the initial functional verification, spanning from the launch phase until the transition of the attitude control system to the regular operation mode. This phase begins with the satellite separation from the launch vehicle. Approximately 4 minutes later, the solar array paddles are deployed. After a further ten hours, the spacecraft shifts to Earth-pointing mode, with the entire critical phase concluding approximately 28 hours after the launch [5].

3. Initial Functional Verification

Following the critical phase, the initial functional verification phase begins, lasting approximately three months. This period is dedicated to verifying the performance of the satellite and its onboard systems, including the primary observation sensors. During this phase, the first sets of data will be acquired by the Thermal And Near infrared Sensor for carbon Observation (TANSO)-Fourier Transform Spectrometer (FTS) and TANSO-Cloud and Aerosol Imager (CAI) sensors and subsequently transmitted to the ground station to ensure full operational integrity.

4. Initial Calibration and Validation

Following the initial verification, the calibration of the mounted sensors and the tuning of the computer processing systems will be conducted by JAXA and the NIES. During this stage, a preliminary analysis of clear-sky carbon dioxide (CO_2) and methane (CH_4) column-averaged mole fractions will be performed both by GOSAT and by ground equipment to validate the retrieval algorithms and ensure data accuracy.

5. Nominal Operation

The nominal operation phase is scheduled for a duration of five years. During this period, the spacecraft performs periodic ConOps on a daily basis, as illustrated in Figure 4. Scientific data is primarily transmitted to high-latitude ground stations such as Svalbard Satellite Station (SvalSat) which can provide frequent transmission windows for the polar-orbiting satellite. Orbital maneuvers are computed onboard and subsequently validated by ground control before execution. Scientific instrument operations are performed primarily on an orbit-length scale, as detailed in Figure 5.

6. Extended Operation and disposal

Given the high reliability and the projected propellant margins of the spacecraft, the mission duration may be extended beyond the nominal five-year period. This phase is subject to the continued functional integrity of the onboard instruments and the approval of extended flight operations by the ground segment. Should either condition no longer be met, the spacecraft will be decommissioned.

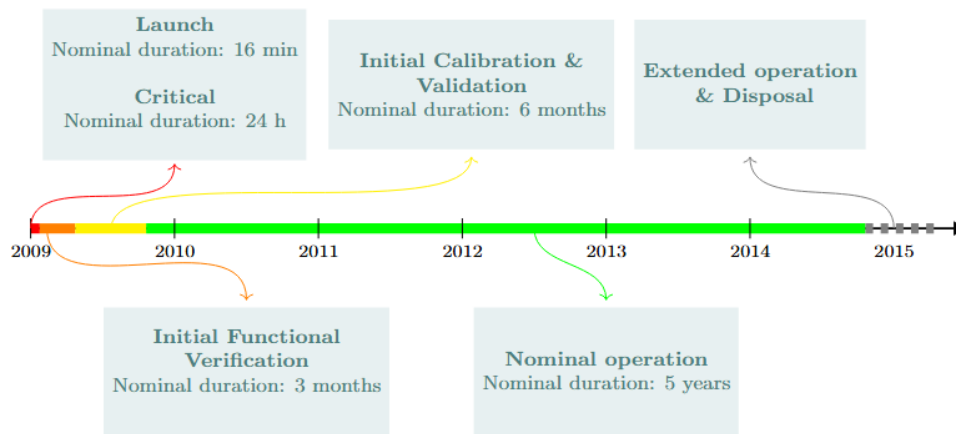


Figure 3: Mission Phases Timeline [1, 5]

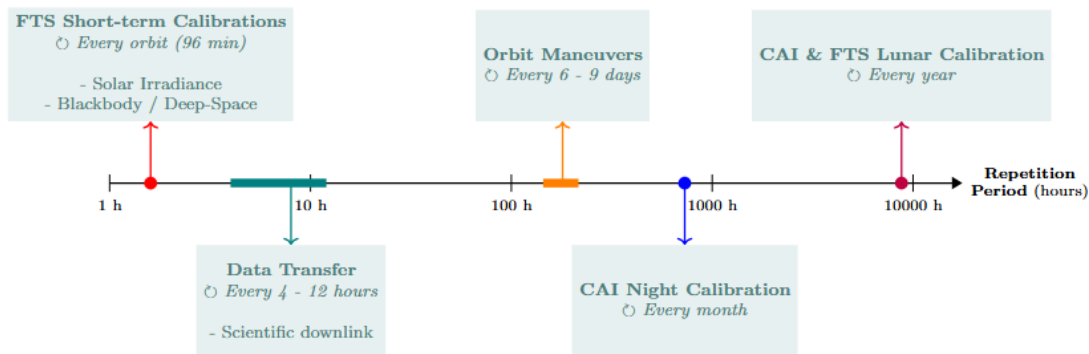


Figure 4: Periodic ConOps [1, 3]

1.7 Scientific Payload

To accomplish the mission objectives, GOSAT carries several scientific payloads, among which the TANSO system serves as the primary instrument.

TANSO-FTS

FTS is a high-resolution Fourier Transform Spectrometer designed to measure the concentration of greenhouse gases such as CH_4 and CO_2 in the Earth's atmosphere by analyzing the absorption features within certain spectral bands [3]. In particular, FTS observes reflected solar radiation in the Short Wavelength InfraRed (SWIR) region and thermal emission in the Thermal InfraRed (TIR) region. Through SWIR observations, it can monitor the column density of greenhouse gases during daytime and primarily over land regions; instead, TIR observations allow the monitoring during nighttime and over the sea.

TANSO-CAI

CAI is a radiometer of Ultraviolet (UV), visible and SWIR, used to detect clouds and aerosols that could interfere with the measurements from FTS. It provides contextual information on the atmospheric conditions surrounding the observation area by acquiring wide-swath images along the satellite's orbit [3]. This data significantly improves the reliability and precision of the FTS measurements through the filtering of contaminated observations before analyzing greenhouse gases concentration.

To better illustrate the operation of the TANSO system, Figure 5 shows how scientific ConOps are organized over the course of a single Earth observation orbit.

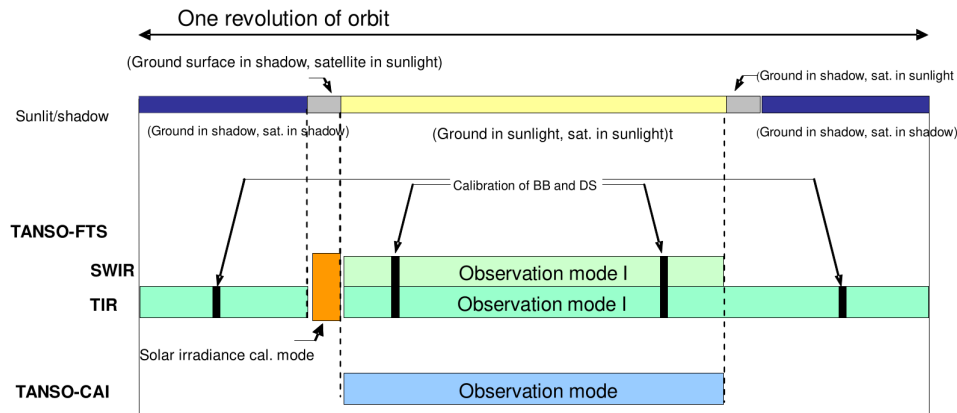


Figure 5: Example of routine ConOps with no sunglint (above land) observation pattern. (BB: Blackbody, DS: Deep-Space) [3]

Technical Data Acquisition Equipment (TEDA)

TEDA is designed to measure the radiation and charged-particle environment encountered by the spacecraft in LEO [6]. The instrument consists of four Light Particle Telescope (LPT) and one Heavy Ion Telescope (HIT) that together measure the flux and energy of electrons, protons, alpha particles, and heavier ions. The collected data is used to better characterize the space radiation environment, which is essential for improving future Earth observing satellite designs.

1.8 Preliminary Trajectory Design and Correlation with Functionalities

A Sun Synchronous Orbit (SSO) is fundamental to achieving GOSAT's scientific and operational objectives. The SSO ensures that measurements are consistently taken under similar lighting conditions (13 : 00 local solar time), optimizing the observation of greenhouse gases by minimizing shadows and maximizing the signal-to-noise ratio for the TANSO-FTS and TANSO-CAI instruments. This consistency is critical for accurate, long-term data comparison and validation.

The high inclination of the SSO not only guarantees consistent lighting but also enables frequent, reliable downlink opportunities at high-latitude ground stations, such as SvalSat. This enables the use of daily data transfer windows, ensuring timely transmission of scientific data during the nominal operation phase. The altitude (~ 666 km) is selected to balance payload performance providing sufficient spatial resolution (100-1000 km) while remaining in LEO. It implies periodic orbit maintenance maneuvers to counteract atmospheric drag and maintain precision in ground track recurrence.

The orbit's repeating ground track (with a recurrence period of 3 days) ensures systematic global coverage, allowing GOSAT to revisit the same regions periodically. This repetition is necessary for tracking temporal changes in greenhouse gas concentrations while ensuring the ground track covers a broad surface area for comprehensive global mapping. Additionally, the systems' design accounts for operation in both sunlit and eclipse phases, directly influencing the thermal control and power subsystem requirements. During eclipse phases, the spacecraft relies on battery storage, while thermal systems must manage temperature fluctuations to protect sensitive instruments.

Finally, the trajectory's compatibility with the H-IIA launch vehicle and launch station dictates constraints on altitude and inclination, to ensure direct injection into the operational orbit. [3]

2 Mission Analysis

2.1 Orbit Design and Maintenance

The GOSAT Spacecraft (S/C) operates in SSO at an altitude of approximately 666 km and an inclination of about 98 degrees. The orbit altitude is designed considering a sub-recurrent ground track with a 3-day repeat cycle according to Figure 6. This configuration ensures a fixed local solar time at the descending node, providing consistent illumination conditions for observations and repeated passage over ground stations.

Additionally, it's worth noticing that the GOSAT mission doesn't require any orbital transfer because it is injected directly into its operational orbit, thus the ΔV budget for this mission includes only initial corrections, station-keeping, and de-orbiting costs.

Injection Correction

Following separation from the H-IIA launcher, the S/C is inserted into an orbit affected by launcher's inaccuracy of ± 10 km on the semi-major axis and $\pm 0.18^\circ$ on the inclination, quantified with 3-sigma precision as reported by [7]. Therefore, initial correction maneuvers are necessary to adjust the orbital elements and can be estimated by calculating the ΔV required for a Hohmann transfer (5.35 m/s), with an additional inclination change (23.64 m/s), considering the worst possible release. It has to be noticed that the insertion uncertainty are given for a 800 km SSO, but since the altitude and inclination of GOSAT are very similar, they are assumed not to change.

Altitude Maintenance

Atmospheric drag causes a continuous decay of the orbit, reducing altitude and altering the orbital period. This effect leads to a drift in the ground track, directly impacting the repeat cycle requirement. To counteract this perturbation, frequent in-plane maneuvers are performed every 6-9 days [3]. The yearly average ΔV has been estimated using Fig. 10-17 in [8, 9], which correlates altitude and ballistic coefficient ($B = 23.6 \text{ kg/m}^2$)¹ with the ΔV cost required to counteract orbital decay, resulting in 2.09 m/s per year. Since this estimate does not involve an exact computation of the maneuvers, 100% ΔV margin for stochastic maneuvers shall be applied.

Furthermore, by changing the semi-major axis, it is also possible to control the RAAN drift rate $\dot{\Omega}$ without performing an out-of-plane maneuver, hence saving a significant amount of ΔV when controlling the position of the descending node.

Inclination Maintenance

Gravitational perturbations, including effects such as Earth's oblateness and third bodies, cause gradual variations in inclination of the orbital plane. Although these effects are slower than drag-induced decay, they must be corrected to ensure stable observation conditions. Inclination correction maneuvers are performed rarely, but require relatively high ΔV due to their out-of-plane nature. In the absence of direct data for GOSAT, a correction interval of 2.5 years is assumed based on similar missions within the GOSAT program [10], effectively resulting in a single correction maneuver over the mission lifetime. Through the simulation of the perturbed orbit², the ΔV required for a single correction halfway through the GOSAT operational period can be computed, corresponding to 6.79 m/s.

De-orbit Maneuver

At the end of the mission, GOSAT must comply with space debris mitigation guidelines. This is achieved by performing a maneuver to lower the perigee altitude to approximately 600 km or below,

¹ $B = m/(A \cdot C_d)$ assuming $C_d = 2.5$ and considering $m = 1750 \text{ kg}$, $A = 29.6 \text{ m}^2$

²Computed integrating Gauss' equations, considering J2, Sun and Moon third body effect

which ensures reentry within 25 years by natural orbit decay due to increase of atmospheric drag [11]. Modelling it as a deterministic impulsive maneuver, the resultant cost is 17.75 m/s, with a 10 m/s margin added according to [12], as it is the only maneuver treated this way.

Reserve

In order to account for other maneuvers such as collision avoidance and to eventually extend the GOSAT lifetime, a reserve of 25% of the usable propellant (after margins) amounting to 22.8 m/s has been added as stated in [13] for preliminary mission analysis.

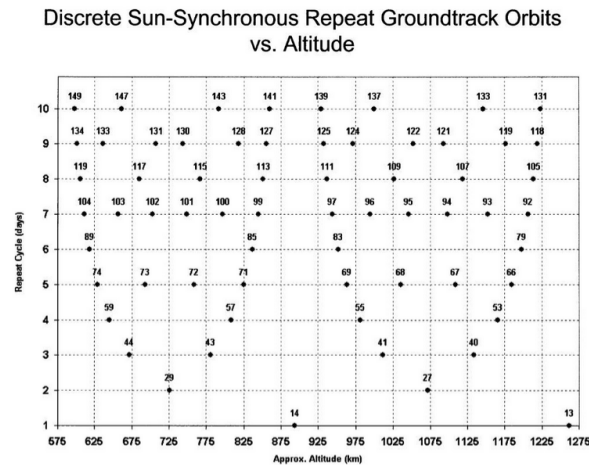


Figure 6: Sun-synchronous repeating track orbits. [14]

2.2 ΔV Breakdown

Maneuver	Literature [m/s]	Computed [m/s]	Margin	After Margin [m/s]
Insertion Correction	[-]	28.99	+ 0 %	28.99
Altitude Maintenance	[-]	10.43	+ 100 %	20.86
Inclination Maintenance	[-]	6.79	+ 100 %	13.58
Reserve (25% of usable propellant)	[-]	15.99	[-]	22.8
De-orbit	[-]	17.75	+ 10 m/s	27.75
Total ΔV	146.70	79.95	[-]	113.98

Table 2: ΔV Breakdown

The literature ΔV can be computed through the Tsiolkovsky equation considering the wet mass (1750 kg) [3], propellant mass (115 kg) [15] and hydrazine's specific impulse $I_s = 220$ s.

The reconstructed ΔV , after the addition of margins, remains lower than the value obtained from literature for several reasons, including the possibility of a larger reserve, given that GOSAT is still operational after more than 15 years. Additionally also maneuvers not considered in this preliminary analysis, such as the use of thrusters for attitude control, contribute to this difference.

3 Propulsion System

Throughout the mission, GOSAT's orbit will gradually drift due to natural environmental effects. The Propulsion System (PS) provides the capability to maintain the intended trajectory, execute the initial injection-correction maneuver, and perform the final de-orbit burn.

As stated in the Mission Analysis section, GOSAT does not need a dedicated primary propulsion engine, as it is directly injected into its target orbit by the launcher and it stays in LEO for its entire mission. Instead, it relies on a unified Reaction Control System (RCS), or secondary propulsion system, using 1N and 20N thrusters to perform both orbit maintenance and attitude control functions.

3.1 Propulsion Architecture

The S/C is equipped with a monopropellant hydrazine blow-down system, the architecture of which is shown in Figure 7. In particular, it includes two 550mm diameter tanks (labeled TNK-1 and TNK-2), four 20N thrusters, eight 1N thrusters, tubes, pressure sensors (PRE-1 and PRE-2), filters, fill and drain valves (FDV), and latch valves (LAV).

The PS is arranged in two identical branches to provide full redundancy in case of a malfunction, a design made possible by the cross-over configuration upstream of latch valves 1-4. This arrangement allows all thrusters to be supplied from either tank; in the event of a leak, isolating the affected tank preserves full propulsion capability through the remaining one.

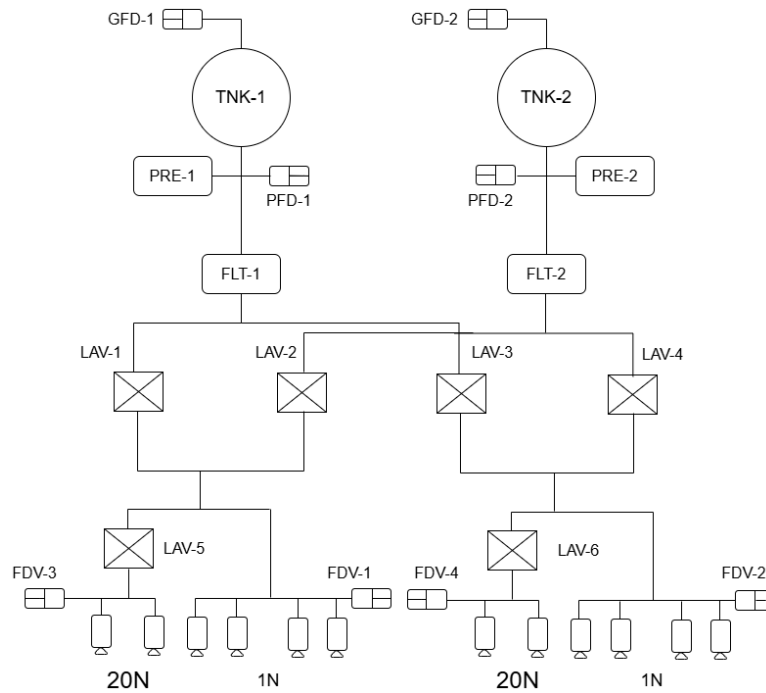


Figure 7: Propulsion System architecture [16]

Monopropellant thrusters

GOSAT uses the MT-9 model for the 1N thrusters and the MT-2 model for the 20N thrusters³, the beginning-of-life characteristics of which are presented in Table 3 [15]. These thrusters use hydrazine as a monopropellant, which decomposes as it passes through a catalytic bed and releases heat.

³both from IHI Aerospace

PS Hardware [16]			Thrusters Performance [17]		
Propellant tanks	n_{tank}	2	20N Pressure[bar]	$p_{inlet,20N}$	4.8–27.6
20N thrusters	n_{20N}	4	20N Thrust [N]	T_{20N}	6.9–19.6
1N thrusters	n_{1N}	8	20N I_{sp} [s]	$I_{sp,20N}$	210–226
Latching valves	n_{LAV}	6	1N Pressure [bar]	$p_{inlet,1N}$	5.5–27.6
Main filters	n_{FLT}	2	1N Thrust [N]	T_{1N}	0.29–1.13
			1N I_{sp} [s]	$I_{sp,1N}$	208–215

Table 3: RCS data

Tanks

GOSAT carries two spherical tanks with an internal diameter of 55 cm and a capacity of 86 L each, both of them containing fuel and pressurant gas. They are made of Ti-6Al-4V titanium and they feature surface tension channel meshes [18]. This latter device ensures that only liquid hydrazine enters the propellant lines.

Geometrical configuration

Satellite's centre of mass is not corresponding to the axis of the main structural carbon-fibre tube primarily due to the payload configuration. As a result, the thrusters are arranged to be approximately symmetrical with respect to the centre of mass rather than the structural geometry [19, 20].

Eight attitude control, 1N thrusters are placed in pairs in proximity of the body corners of the bottom plate and canted with respect to X axis. This configuration allows for maximum torque on the S/C as well as shields the payload from the exhaust gases. Each pair is composed of thrusters from the two separate propulsion subsystems.

The offset between the propellant tanks and the satellite's centre of mass causes the centre of mass to move over time as the propellant is depleted. Since that, the four 20N thrusters are placed in a way to allow for the attitude control during the orbital maneuver burns - not in the midpoint of the main plate but spread around allowing for torque control arm. They are activated in diagonal pairs, from subsequent propulsion subsystems.

Tanks are placed in the aft section of the satellite, along the X-axis due to the dimensional constraints and the proximity to the thrusters.

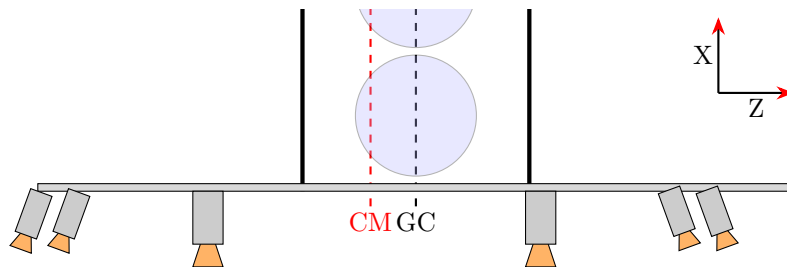


Figure 8: Simplified diagram of thruster configuration. Where, CM - centre of mass and GC - geometrical centre (structural tube).

3.2 Propulsion Design Choices

Propulsion type. First of all, the simplicity of monopropellant pressure-fed control systems makes them highly desirable for S/C attitude and trajectory control applications. This not only leads to

lower costs and reduced weight but also to a lower risk of failure, since monopropellant thrusters do not require an ignition system and rely on catalytic decomposition.

This architecture allows for repeated low- ΔV maneuvers over a 5-year operational period, and since the propulsion system lifetime is typically dictated by the catalyst bed, its longer lifetime ensures compatibility with the mission requirements. A monopropellant RCS is therefore the most suitable compromise between simplicity, reliability, restart capability, and sufficient maneuvering performance.

Propellant. The propellant commonly used for this type of propulsion is hydrazine, a colorless and toxic compound that decomposes in the presence of a catalyst. It is stable for years even when subjected to mechanical shocks, which is in accordance with the mission's lifetime requirement, and stays liquid at room temperature, which simplifies storage compared to cryogenic propellants. Finally, its low ignition time reduces the delay associated with valves opening. Its use over several decades makes it one of the most reliable fuels for AOCS propulsion.

Thrusters. The supplier of the thrusters, IHI Aerospace, is the same as for many missions developed by JAXA, and their reliability has been proven [15]. IHI provides thrusters in 1N, 4N, 20N, and 50N classes, all considered for initial selection.

A single, simple feed and pressure system is required to supply both orbital and attitude thrusters at similar operating pressures. The 50N thruster has a minimum operating pressure of 0.9MPa, while the other thrusters operate around 0.5MPa. Considering system simplicity and weight constraints, the 20N thruster is chosen as the baseline option. Using two 20N thrusters in a quasi-impulsive maneuver (approximately 1% of the orbital period), the semi-major axis can be adjusted by 2.5km and the inclination by 0.01° , which meets the mission requirements. Attitude control is provided by 1N thrusters for de-tumbling and reaction wheel desaturation. For an initial angular rotation rate of $2^\circ/\text{s}$ [7] around the minimum inertia axis, the thrusters are capable of de-tumbling the S/C in approximately 6 minutes⁴. Each thruster incorporates catalyst bed and valve heaters to maintain proper operating temperatures, ensuring reliable hydrazine decomposition while preventing catalyst damage, propellant freezing, and valve seat degradation.

Tanks. From a structural efficiency point of view, spherical tanks are optimal for propellant storage. They achieve the highest volume-to-mass ratio, minimizing the overall weight of the PS, and provide optimal stresses distribution. It is also worth noting that titanium is chosen due to its compatibility with hydrazine [21] and its structural properties. Additionally, to maintain the hydrazine within its operational temperature range, thermal insulation foil is applied to the system. The choice of two tanks instead of a single one, which would theoretically be enough for this type of propulsion, is justified not only by storage capacity, but also by redundancy requirements. Since no single failure shall compromise the complete propulsion capability of the S/C, the dual-tank architecture improves robustness in case of leakage or branch isolation.

Pressurization system. A blowdown pressurization system is particularly suitable for GOSAT because the S/C requires a limited total ΔV , distributed over many relatively small maneuvers during the mission lifetime, rather than long continuous burns. The absence of active pressure regulation hardware significantly simplifies the system and improves reliability. Additionally, N_2 is selected over He as pressurant regardless of its higher molecular weight due to its lower leakage rate. This architecture is therefore fully consistent with the mission profile and the overall design objective of minimizing subsystem complexity.

⁴Computed assuming constant torque provided by two 1N thrusters, inertia of a rectangular cuboid and thrusters located at edges of S/C.

3.3 Propulsion Sizing

Data

Propellant, Pressurant and Material ⁵					
System wet mass [kg]	M_{wet}	1750	N ₂ H ₄ density [kg/m ³]	ρ_{prop}	1010
Propellant mass [kg]	M_{prop}	115	Ti-6Al-4V ultimate stress [MPa]	σ_{Ti}	950
Pressurant $R_{specific}$ [J/(kg · K)]	R_{N_2}	296.8	Ti-6Al-4V density [kg/m ³]	ρ_{Ti}	4512

Table 4: Propellant, pressurant and material data

3.3.1 Propellant sizing

Since a verified ΔV budget was unavailable in the literature, the initial sizing was based on the system's known wet mass and propellant mass. Using these parameters, the required propellant volume was calculated and a 10% unusable volume margin was applied in accordance with [12].

$$V_{prop} = \frac{M_{prop}}{\rho_{prop}} = 0.114 \text{ m}^3 \quad V_{prop,real} = V_{prop} \cdot 1.1 = 0.125 \text{ m}^3 \quad (1)$$

3.3.2 Pressurant sizing

Since the blowdown ratio and pressurant pressure were not directly available, they were estimated based on the thruster inlet pressure (p_{inlet}). The operating ranges of both the 1N and 20N thrusters were analyzed to determine the system's pressure limits. While both thrusters share the same maximum operating pressure, the 1N thruster has a higher minimum pressure requirement, which established the lower bound for the system. Based on this analysis, the design limits were set at $p_{inlet,max} = 27.6$ bar and $p_{inlet,min} = 5.5$ bar. The pressurant pressure profile was derived by applying pressure loss corrections to the inlet limits. Specifically, a static pressure loss of $\Delta p_{feed} = 0.5$ bar was allocated for the feeding lines, and a dynamic pressure drop of $\Delta p_{dyn} = \frac{1}{2} \rho_{prop} v_{prop}^2 = 0.505$ bar was evaluated based on a mean propellant velocity of 10 m/s. These contributions were added to the p_{inlet} values to determine the final required pressurant pressures under Δp_{loss} term.

$$p_{gas,i} = p_{inlet,max} + \Delta p_{loss} = 28.6 \text{ bar} \quad p_{gas,f} = p_{inlet,min} + \Delta p_{loss} = 6.5 \text{ bar} \quad (2)$$

With the initial and final pressurant pressures established, the blowdown ratio was determined. This ratio was subsequently used to calculate the required initial pressurant volume under isothermal assumption.

$$B = \frac{p_{gas,i}}{p_{gas,f}} = 4.40 \quad V_{gas,i} = \frac{V_{prop,real}}{B - 1} = 0.037 \text{ m}^3 \quad (3)$$

With the initial pressure and volume of the pressurant established, the gas mass was calculated assuming ideal gas behavior at a constant tank temperature of 293 K. Under steady-state conditions, the tank pressure is assumed to be in equilibrium with the gas pressure ($p_{tank,i} = p_{gas,i}$). Finally, a 20% margin was applied to the resulting gas mass in compliance with ECSS standards (MAR-MAS-090).

$$M_{gas} = \frac{p_{tank,i} \cdot V_{gas,i}}{R_{N_2} \cdot T_{tank}} = 1.213 \text{ kg} \quad M_{gas,real} = M_{gas} \cdot 1.2 = 1.455 \text{ kg} \quad (4)$$

⁵Material properties are listed at 293 K

3.3.3 Tank sizing

Since $V_{prop,real}$ and $V_{gas,i}$ are both known, the tank volume can be then evaluated and then the real volume is obtained taking into consideration the volume occupied by the surface tension system and so a 1% margin on the volume was considered.

$$V_{tank} = V_{prop,real} + V_{gas,i} = 0.1621 m^3 \quad V_{tank,real} = V_{tank} \cdot 1.01 = 0.1637 m^3 \quad V_{single,tank} = 0.0819 m^3 \quad (5)$$

Due to the dual-tank configuration, the total required volume was divided equally to characterize a single tank. From this allocated volume, the tank radius r_{tank} was derived. A burst pressure was then defined as $p_{burst} = 2 \cdot p_{gas,i}$, which served as the basis for calculating the wall thickness t_{tank} of each tank.

$$r_{tank} = \left(\frac{3 \cdot V_{single,tank}}{4 \cdot \pi} \right)^{\frac{1}{3}} = 269.4 mm \quad t_{tank} = \frac{p_{burst} \cdot r_{tank}}{2 \cdot \sigma_{Ti}} = 0.811 mm \quad (6)$$

The final diameter obtained for a single tank is 538.8 mm, which leads to the selection of the 550 mm option, as it is the closest larger size available from IHI.

With the wall thickness established, the mass of a single tank was calculated. This allowed for the determination of the total structural mass for the dual-tank configuration, concluding the propellant system sizing.

$$M_{single,tank} = \frac{4}{3} \cdot \pi \cdot \rho_{Ti} \cdot ((r_{tank} + t_{tank})^3 - r_{tank}^3) = 3.347 kg \quad (7)$$

3.3.4 Mass budget

The total mass budget for the monopropellant system (M_{PS}) is defined as the sum of the tank structure, the pressurant gas, and the thruster assemblies. Thruster masses were sourced from the MOOG commercial catalogue [22] as IHI mass data was not available; notably, the mass of the MOOG 22N thruster was utilized as the baseline for the 20N thruster requirement. The final mass of the PS system is obtained by adding a 10% margin for cables and harness.

$$M_{PS} = 2 \cdot M_{single,tank} + M_{gas,real} + 4 \cdot M_{20N} + 8 \cdot M_{1N} = 14.069 kg \quad M_{PS,real} = M_{PS} \cdot 1.1 = 15.476 kg \quad (8)$$

3.3.5 Power budget

Power consumption is divided between the 1N and 20N thruster subsystems, as summarized in Table 5. Solenoid valves are provided by MOOG [23]; their power usage is primarily associated with heating to the operational temperature range (5°C - 121 °C)[24] and actuation during opening.

The power requirements of the catalyst bed heaters are estimated based on laboratory tests and datasheets from comparable thruster models [25, 26]. Their temperature is raised to approx. 200°C and 30°C for 1N and 20N thrusters correspondingly [24].

	1 N thrusters ×4	20N thrusters ×2
Valves [W]	3.8×4	20.5×2
Catalyst heatbed [W]	4×4	12×2
Total in subsystem [W]	31.2	65

Table 5: Propulsion power budget

4 TMTC Architecture

4.1 Overview

The TMTC subsystem provides the communication link between the spacecraft and the ground segment, supporting downlink transmission of scientific and housekeeping data, uplink of commands, and tracking functions. In particular the TMTC subsystem of GOSAT employs an S-band system for tracking and command operations and an X-band link for the transmission of scientific data.

4.2 Space Segment Architecture

Within the Space Segment, the analysis of the communication subsystem begins with the logical architecture defined in the official system block diagram [27]. The TMTC subsystem in S-band is designed with three distinct antennas (SANT-1, 2, and 3) to accommodate two fundamentally different operational scenarios. During nominal operations, when the satellite maintains a stable Nadir orientation, the system relies on a primary antenna optimized for Earth-coverage. Rather than requiring strict directional pointing, this antenna, with cylindrical geometry as seen on the Nadir-facing panel [28], is designed with a broad emission profile that efficiently illuminates the entire visible Earth disc.

To ensure continuous communication even in the event of attitude loss (e.g., during launch phases or Safe Mode tumbling), the TMTC architecture must provide near-omnidirectional coverage. This is achieved by employing a hybrid coupler (SHYB) that drives the two remaining S-band antennas (SANT-2 and SANT-3). As established in aerospace engineering literature, combining two hemispherical antennas on opposite faces through a coupler is a standard architecture to eliminate blind spots caused by the spacecraft's main body and guarantee full spatial connectivity in any direction [29]. One of these antennas (SANT-2) can be identified on the front face of the satellite [30], while the opposite antenna's (SANT-3) location on the thruster panel was confirmed by the protective cover visible in [31].

Parallel to this architecture, the high-speed downlink of scientific telemetry is managed by the Direct Transmission (DT) subsystem. System schematics reveal that observation data generated by the primary instruments is routed to the Mission Data Processor (MDP), which incorporates a 48 GB Solid State Recorder (SSR) divided in three 16 GB drives [32, 3]. This substantial storage capacity is a critical requirement derived from the mission's data budget. Once the spacecraft establishes a line-of-sight with high-latitude receiving facilities or domestic stations, the accumulated data is rapidly downlinked via X-band. To support this high-capacity link, the architecture utilizes two redundant X-band antennas (XANT-A and XANT-B). Because this high-speed downlink occurs exclusively during nominal Earth-pointing operations, these hemispherical antennas are rigidly mounted on the Nadir-facing panel, physically adjacent to the primary SANT-1 antenna. [28].

4.3 Ground Segment

The ground segment is the interface between Space and Earth and it includes both the tracking stations (antennas) and the Mission Control Center. It ensures the reception, processing, and archiving of data, while monitoring the satellite's health, attitude, and trajectory in real time. As the mission's central decision making entity, the ground segment is responsible for generating and transmitting the command sequences required for mission operations and anomaly management.

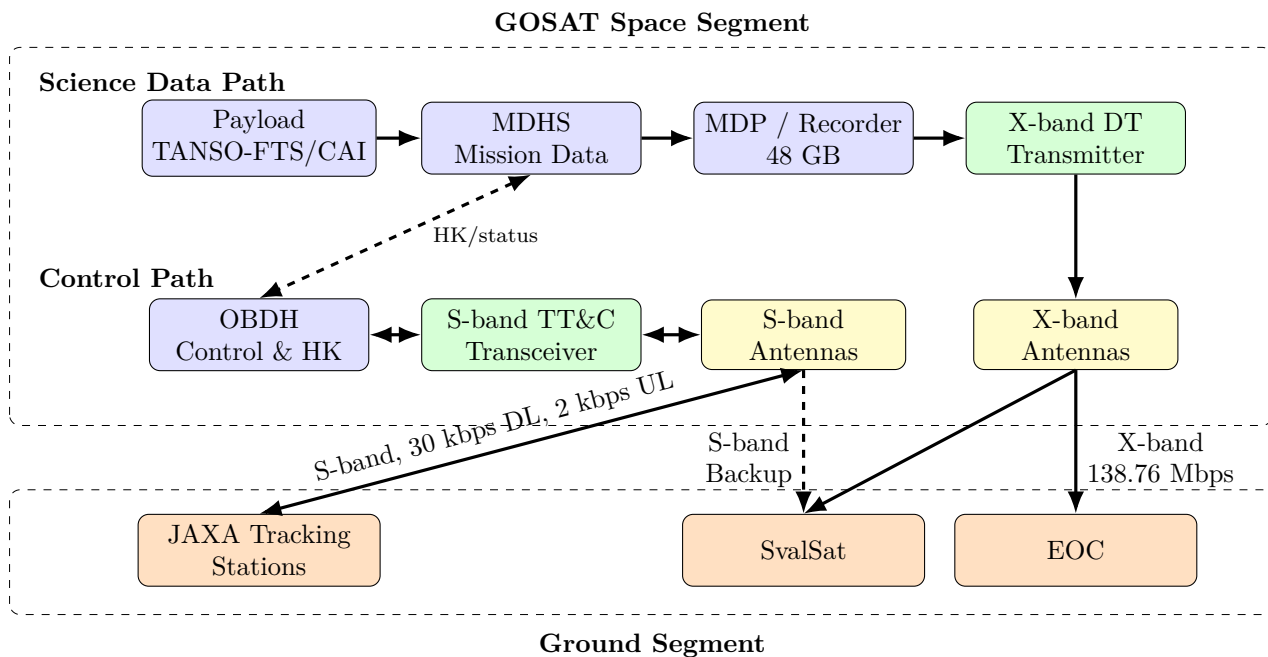


Figure 9: TMTC subsystem architecture of GOSAT.

The low data rate and exchange frequency requirements for satellite telemetry and commands allow JAXA's global tracking stations to be sufficient on their own. The use of the S-band is justified by its robust, low-noise signal and its minimal atmospheric attenuation. Furthermore, it requires ground reception equipment that is highly cost-effective and compact, making it ideal for low data rate transmissions.

GOSAT naturally uses JAXA's Earth Observation Center (EOC) in Hatoyama to transmit payload data. It consists of X-band antennas, a data storage and transmission system, as well as a planning and management system for the connections with all the satellites it supports. However, its location is not optimal as it is not regularly overflown by the satellite. Consequently, GOSAT also transmits its payload data via the Svalbard station. This choice was made considering the high-latitude location (78°N) of the station, which is very convenient for GOSAT considering its SSO orbit. It allows the antennas to maintain visibility of the satellite during almost every revolution, making it ideal for downloading large volumes of payload data. As a backup, SvalSat can also be used for S-band data transmission.

All received data is then transmitted to the mission control center, the Tsukuba Space Center, which itself is composed of various specialized entities. The JAXA tracking stations are directly linked to the Satellite Control System, which monitors the spacecraft's health, orientation, and trajectory. This system is responsible for processing and analyzing housekeeping telemetry. Furthermore, in close coordination with the Flight Dynamics System (FDS), the Satellite Control System (SCS) also manages and transmits the commands generated by the latter to the satellite. Regarding the payload data received by Svalbard and the EOC, it is then transmitted to the EORC Information System (EIS), which stores and manages all technical data. The majority of this data is eventually processed by the Global Earth Observation Data Processing and Control System, making it usable and readable. The control center also houses the Mission Operation Management Organization (MMO) System, which receives user requests, prioritizes them, and transmits the resulting commands to the SCS.

4.4 TMTC Operations & Phases Correlation

Critical Phase

During this phase, as stated in subsection 4.2, an omnidirectional communication channel is required; therefore, GOSAT relies on S-band telemetry. Although the S/C primarily operates using stored commands, automatically executed at predefined times, essential initial functional checks, such as equipment startup, are carried out via real-time command execution and telemetry monitoring while the satellite is in view of the ground stations [otsumoto_ibuki].

Initial Functional Verification

In the initial functional verification phase, it is necessary to verify the integrity of the satellite and to play back and downlink all telemetry recorded in the memory. Therefore, similarly to nominal operations, downlink sessions are performed once every two orbits at the JAXA and SvalSat stations to acquire mission data, including housekeeping information, via X-band [otsumoto_ibuki].

Calibration, Validation and Nominal Operation

Once the initial check-up is completed and GOSAT enters the calibration and subsequent nominal operations phase, it will start downlinking at SvalSat or EOC approximately once every two orbits via X-band, transferring ≈ 200 minutes of TANSO measurements to be sent to Tsukuba for processing. While mission data and satellite health monitoring (HK telemetry) are primarily handled via X-band for operational efficiency, a global network of seven S-band stations is maintained for constant tracking and command transmission. In this architecture, the Svalbard station also serves as a backup for S-band telemetry and command operations specifically during satellite anomalies [33].

4.5 Data Volume and Frequency Selection

The payload data and the housekeeping telemetry data make up the downlink, while the commands sent from the ground segment to the satellite make up the uplink, which requires high reliability despite the relatively low data rate of 2 kbit/s [1]. This low rate ensures a high energy per bit to noise density ratio (E_b/N_0), guaranteeing reliable command transmissions even in degraded link conditions. Housekeeping telemetry is sent at 30 kbit/s [1]; as for the mission data, the scientific payload generates a large amount of data, thus requiring a higher downlink data rate of 138.76 Mbit/s [27].

The S/C has between 8 to 10 daily downlink opportunities to transmit stored data : 6 times a day to SvalSat and between 2 and 4 times a day to EOC [3]. As computed in subsubsection 5.3.1, the average contact time per pass over a repeating-track period is 8.22 minutes. Assuming a constant downlink rate, the average data volume transmitted per pass is 8.55 GB for the Svalbard station, which corresponds to approximately half of the capacity of one of the 3 drives on the satellite.

The choice of frequency band depends on the volume of data to be sent and also on the allocation of frequency bands by International Telecommunication Union (ITU). GOSAT uses a frequency around 2 GHz [33] for telemetry data and 8 GHz for scientific data [1]. These choices are in accordance with ITU, which specifies that telemetry communications have priority on the [2,025-2,110 MHz] band for downlink and [2,200-2,290 MHz] for uplink, while scientific transmissions have priority on the [8,025-8,400 MHz] band [34].

5 TMTC Sizing

This section presents the sizing and performance analysis of the TMTC communication subsystem, including link budget calculations, hardware selection, and operational constraints for both X-band

and S-band links.

	f [GHz]	R_{net} [kbps]	D_{rx} [m]	GT_{rx} [dB/K]	T_{sys} [K]	η_{rx} [-]	P_{tx} [W]	BER_m [-]
X_{down}	8	$138.76 \cdot 10^3$	11.3	-	-	0.55	20	10^{-7}
S_{down}	2	30	11.3	22.6	-	0.55	5	10^{-7}
S_{up}	2	2	-	36.8	500	0.55	69	10^{-5}

Table 6: Communication system parameters for X-band and S-band links

Table 6 summarizes the main parameters used for sizing calculations. These include carrier frequency f , net data rate R_{net} as well as T_{sys} for S/C antenna. Ground segment antenna diameter D_{rx} and the receiver gain-to-noise-temperature ratio GT were obtained from article [35]. A constant antenna efficiency $\eta_{rx} = 0.55$ is assumed for all cases, as it is a typical value. The transmitted power P_{tx} represents the RF output level of each link, while the bit error rate BER_m follows ESA standards.

5.1 Signal Manipulation

According to [36], both Reed-Solomon encoding and convolutional encoding are applied, with corresponding code rates $R_{c,RS} = 223/255$ and $R_{c,conv} = 0.5$. For these settings, a rather conservative value of $E_b/N_0 = 5$ dB is selected for the downlink, corresponding to convolutional encoding with a target bit error rate of $BER_m = 10^{-7}$. For the uplink, a lowered target bit error rate of $BER_m = 10^{-5}$ is assumed, leading to a reduced requirement of $E_b/N_0 = 4$ dB. Additionally, OQPSK digital modulation ($m = 2$ bits/symbol) is employed.

5.2 Hardware Selection

X-band antenna: A helical antenna operating in broadside mode is considered [37]. This type of antenna provides hemispherical coverage with a gain between -1 dB and 3 dB over a wide angular range, spanning from 0° to 90° . For this reason, a worst gain value of -1 dB was considered and the pointing losses were not calculated.

S-band antenna: Although the following computation should be carried out for both S-band antenna configurations, only the most critical case, corresponding to the lower-gain dipole antenna, has been considered for both uplink and downlink in this band. In particular, a $\lambda/2$ dipole antenna⁶ placed at a distance of $\lambda/4$ from a reflector has been considered. This shields the antenna from the radiation reflected by the S/C and increases the gain of the antenna to ≈ 3 dB, as reported in Fig. 12-4 of [38].

Solid State Power Amplifier (SSPA): Considering the relatively low power required to establish a ground connection from LEO, SSPAs are selected for both frequency bands, with output powers of 20 W and 5 W and input powers of 98 W and 26 W for X- and S-band, respectively [39, 40].

5.3 Ground Station & Contact Strategy

The receiver ground antenna gain were computed based on values provided in [35]. The ground station performance is characterized by the gain-to-noise-temperature ratio GT_{rx} . The system noise temperature is then obtained as

$$T_{sys} = 10^{\frac{Grx - GT_{rx}}{10}}. \quad (9)$$

⁶where λ is the wavelength

5.3.1 X-Band

GOSAT is designed to downlink scientific data during contact windows typically ranging from 4 to 10 minutes [41]. An elevation angle of 8.1° corresponds to the minimum elevation required to achieve a 10-minute downlink window in the case of a near-overhead pass. For this reason, it is adopted as the minimum elevation threshold. Based on the simulated visibility windows, the average contact time per pass over a repeating-track period is 8.22 minutes.

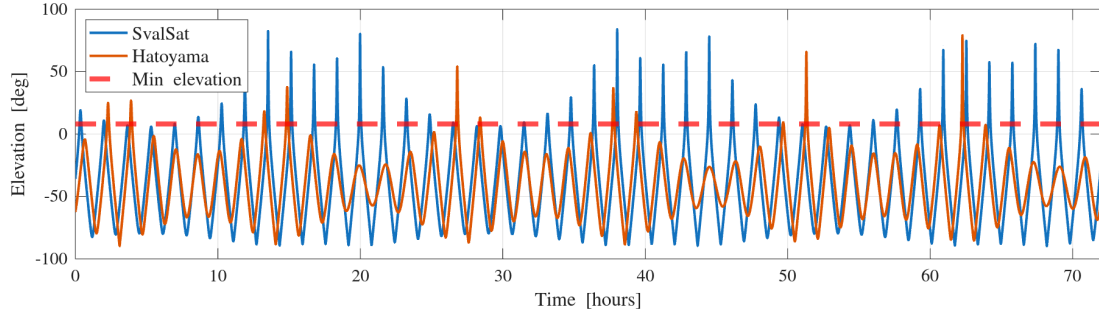


Figure 10: S/C elevation over Svalbard and Hatoyama over one repeating ground-track period.

5.3.2 S-Band

In telemetry command operations, the operational policy is to avoid creating a period of no visibility of more than 4 hours (up to a maximum of 6 hours in the event of a ground station malfunction) [otsumoto_ibuki]. Therefore, ground stations are assigned accordingly Figure 11.

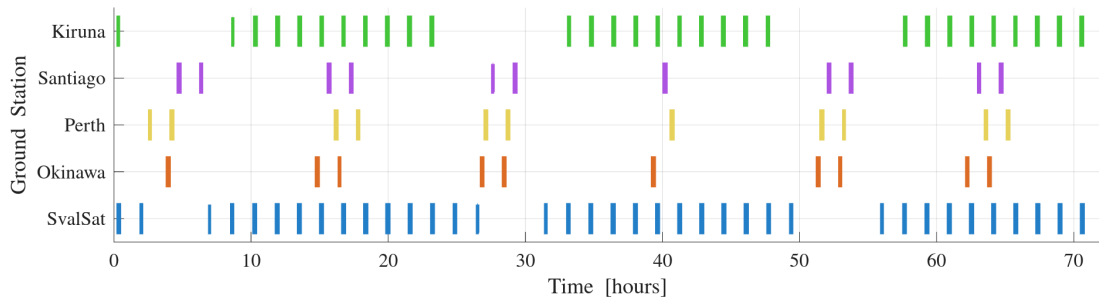


Figure 11: Visibility Windows per Ground Station (lines thickened)

Kiruna is treated as a mandatory ground station, while Santiago, Perth, and Okinawa are not strictly required due to overlapping visibility windows. If Kiruna becomes unavailable, SvalSat is required as a backup, as the remaining stations alone would lead to visibility gaps exceeding 6 hours.

5.4 Link Budget

The link budget quantifies the overall signal power balance of each communication link by accounting for transmission gains, propagation losses, and receiver performance in order to evaluate the resulting signal-to-noise ratio and system margin.

Space losses account for the attenuation of the electromagnetic signal as it propagates through space. They were evaluated based on the signal wavelength and the separation distance d between the ground station and the S/C, with d taken at the minimum elevation condition during contact, which corresponds to the maximum link distance.

Atmospheric loss is estimated using *International Standard Atmosphere* model and *Mean Annual*

Global Reference Atmosphere using *International Telecommunication Union* atmospheric gas attenuation model [42, 43, 44]. The signal trajectory is discretized, and losses are summed over consecutive atmospheric layers while accounting for the minimum elevation of the S/C above the horizon. This approach provides a greater degree of accuracy at low elevations than the commonly used approximation of dividing the zenith attenuation by the sine of the elevation angle.⁷

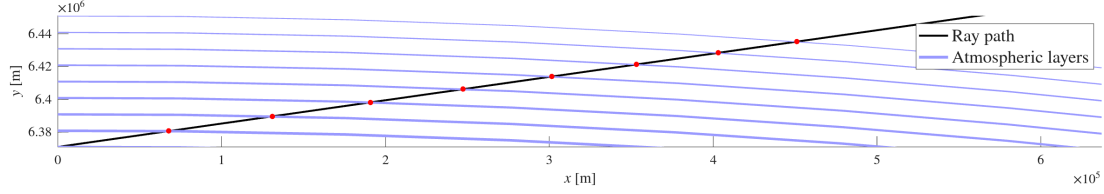


Figure 12: Atmospheric layers- 1/10 computed layers displayed.

Rain loss is estimated based on the rain attenuation curves [45] for a given frequency and worst case elevation angle and presented in Table 7.

Modulation losses are computed for S-band links, as recommended in [46], where modulation index $\beta_{mod} = 57.7^\circ$ is assumed for a residual carrier technique. However, as previously stated, OQPSK modulation is applied to both X-band and S-band links.

Pointing losses are evaluated considering only the ground antenna misalignment $\theta_{err} = 0.1^\circ$. The S/C pointing errors are neglected due to their wide coverage operating mode, and is instead implicitly accounted for by adopting the minimum gain value within the relevant angular range.

Cable losses typically fall within the range of -1 to -3 dB; in this instance, a value of -2 dB is assumed.

	P_{Tx}	G_{Tx}	G_{Rx}	L_{space}	L_{atmo}	L_{rain}	L_{mod}	L_{point}	$EIRP$	P_{Rx}	E_b/N_0	SNR	E_b/N_0 mg	SNR mg
	[dBW]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dB]	[dBW]	[dBW]	[dB]	[dB]	[dB]	[dB]
X_{down}	13.01	-1	56.93	-177.52	-0.31	-2	0	-2.23	10.01	-115.12	9.77	28.61	4.77	25.95
S_{down}	6.99	3	44.89	-165.48	-0.24	-0.02	-1.46	-0.14	7.99	-113	44.92	27.11	39.92	24.45
S_{up}	18.39	44.89	-1	-165.48	-0.24	-0.02	-1.46	-0.14	61.28	-105.46	59.92	31.96	55.78	30.3

Table 7: Link budget parameters for X-band and S-band links (margins in the last column)

The high link margins reflect a conservative design choice that provides additional robustness against long-term system degradation while ensuring reliable performance.

5.5 Mass and Power Budget

The mass and power requirements for the communication subsystem components are summarized in Table 8, based on the selected hardware specifications.

	X-SSPA [39]	S-SSPA [40]	X-ANT [37]	S-ANT1 [47]	S-ANT2 & 3 [48]
Mass [kg]	2.05	1	0.325	0.235	0.1
Power [W]	98	26	[-]	[-]	[-]

Table 8: Mass and Power budget

⁷The model was validated against standard total, dry air, and water-vapour zenith attenuation profiles at sea level, within the resolution limits of the reference data.

6 AOCS Architecture

6.1 Overview

The AOCS subsystem is responsible for determining and controlling the satellite's state vector through a closed-loop architecture. GOSAT's orbit control system allows it to autonomously calculate the control actions necessary to maintain a relatively constant Local Time of Descending Node, Semi-Major Axis, and Eccentricity. Nevertheless, the satellite must wait for approval from the ground teams before performing maneuvers [3]. GOSAT features two attitude determination systems that provide indispensable redundancy for such an observation mission, where the payload requires precise pointing to perform the necessary observations. Specifically, the high-precision system relies on Star Trackers (STTs) and GPS Receivers (GPSRs), while the standard system utilizes Fine Sun Sensors (FSSs) and Earth Sensors (ESs), with the Inertial Reference Unit (IRU) supporting both continuous paths. Their models and specific use will be detailed in the following sections.

Based on data from the sensors, the Attitude and Orbit Control Electronics (AOCE) determines the satellite's attitude and trajectory, compares them with the optimal state vector, and calculates the necessary control actions. The AOCE then transmits commands to the actuators to execute the control maneuvers. These actuators consist of RWs, MTQs, and thrusters, which provide both torque and thrust.

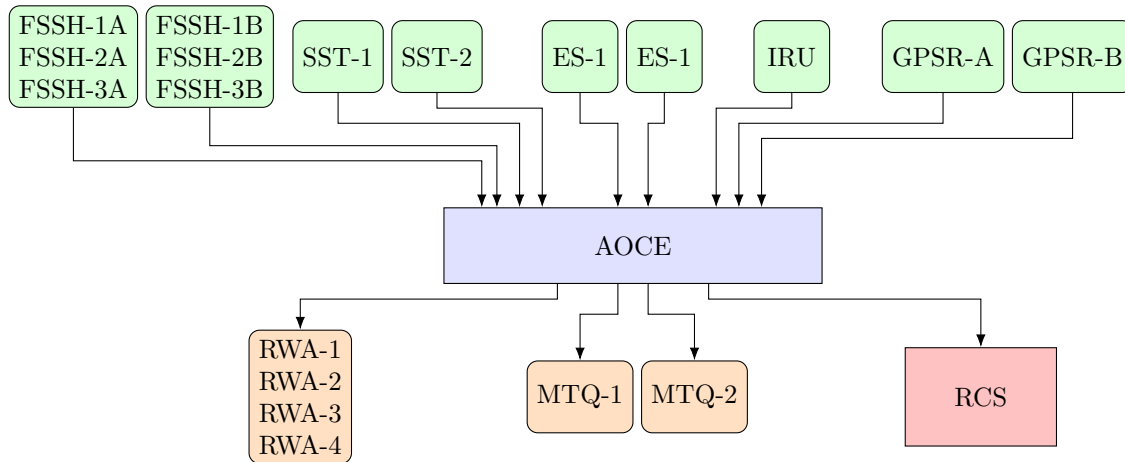


Figure 13: AOCS subsystem scheme

6.2 Control Modes & Pointing Budget

The S/C operates in different control modes, driven by the mission phase and operational requirements.

- The **Detumbling** mode's objective is to dissipate the spacecraft angular rates, enabling sensor measurements for subsequent slew and pointing maneuvers. This mode uses IRU as sensor and 1N-thrusters until the angular velocity is below $0.1^\circ/\text{s}$ [49].
- In **Sun Acquisition** mode, it is sufficient for the S/C to orient its solar panels toward the Sun, thus precise pointing, as well as full attitude determination are not required. Considering also that GOSAT's solar arrays are equipped with a paddling system, a maximum pointing error of 10° is assumed [50]. This mode is initially used during the critical phase immediately after detumbling to ensure that GOSAT maintains sufficient power in the early stages of the mission, and can be later activated in case of anomalies as a safe pointing mode to address non-nominal

conditions. FSSs are naturally the main sensors used for this mode.

- The FSS, ES and IRU are used as sensors in **Standard Attitude Control** mode to maintain a coarse nadir pointing attitude. Due to the absence of a reference value for this mode precision, an Attitude Performance Error (APE) one order of magnitude higher with respect to High Precision Mode is considered, leading to $APE < 0.5^\circ$, which also enables the STT to start its operation safely.
- The nominal mode used for scientific observations is the **High Precision Attitude Control** mode. Once TANSO points toward Earth, the combined use of STT, GPSR and the IRU refine the precision for extremely stable pointing. Considering the spatial resolution of the TANSO system (500 m [3]) and the orbital radius, it's possible to compute the pointing precision required for the observations as $\tan^{-1}(0.5/666) = 0.043^\circ$, which is coherent with the value of 0.04° that can be found in [51].
- The **Moon pointing** mode reorients TANSO, normally directed toward the Earth, toward the Moon to perform annual calibration of the instrument. For this operation the satellite switches from nadir-pointing to inertial navigation using IRU to find the Moon [49]. Considering that the Moon's apparent angular radius is approximately 0.25° and thus much larger than the CAI field of view⁸ (0.043°), this value can be adopted as the effective pointing budget for the mode. A tighter pointing budget of 0.013° is reported in [49], as it represents the actual achieved spacecraft performance rather than the design requirement.
- **Slew** mode is used to reorient the spacecraft between operational attitudes and is characterized by controlled rotational maneuvers with a maximum angular rate limited to $0.5^\circ/\text{s}$, as defined in [9], to prevent excessive structural loads during attitude transitions.

Mode	Attitude target ⁹	Sensors used	Performance
Detumbling	Angular rate damping	IRU	<u>Ang. rate</u> $< 0.1^\circ/\text{s}$ [49]
Sun acquisition	+X axis towards the Sun	FSS, IRU	$APE < 10^\circ$ [50]
Standard attitude	+Z axis towards the Earth	FSS, ES, IRU	$APE < 0.5^\circ$
High precision	+Z axis towards the Earth	STT, IRU, GPSR	<u>$APE < 0.04^\circ$</u> [51] <u>$AKE < 0.03^\circ$</u>
Moon pointing	+Z axis towards the Moon	IRU	<u>$APE < 0.013^\circ$</u> [49]
Slew	Attitude transition maneuver	-	Ang. rate $< 0.5^\circ/\text{s}$ [9]

Table 9: Pointing Budget (underlined for literature values)

6.3 Attitude Determination & Sensors

To physically carry out the attitude and orbit determination functions described in the Overview, the AOCS relies on a highly specialized hardware platform. Because no single unit is self-sufficient across the entire mission profile, the measurements provided by these sensors are mathematically fused, primarily via a Kalman Filter processing during the High-Precision Mode, to ensure precise, real-time state estimation. Furthermore, to eliminate single points of failure and comply with strict Fault Detection, Isolation, and Recovery (FDIR) standards, the hardware is configured as follows [27]:

⁸CAI is considered because FTS has a larger FOV

⁹+Z is the face containing the TANSO system, while +X corresponds to the face nominally oriented in the direction of travel

- **Star Trackers (STT):** Dual configuration (STT-1, STT-2) serving as the primary sensors for high-precision attitude determination [27]. The units feature highly sensitive charge-coupled devices (CCDs) capable of detecting stars down to the 6th or 7th magnitude, coupled with an all-sky onboard star identification software [49]. Flight data confirms a single-STT determination accuracy of approximately 16 arcseconds (0.0045° in Roll, 0.0031° in Pitch, and 0.0033° in Yaw, 3σ) [49]. To ensure robustness against environmental optical disturbances (such as simultaneous Earth interference or blinding), the AOCS is designed to temporarily suspend the high-precision optical filter updates to prevent the ingestion of erroneous data, seamlessly relying on the IRU until the STT field of view is clear [51].
- **Inertial Reference Unit (IRU):** It measures spacecraft angular rates to provide continuous attitude information. The IRU is a critical component for attitude determination, acting as the sole source of attitude knowledge through inertial navigation whenever external optical references, such as the STTs, are temporarily unavailable or blinded by environmental interference [51]. Furthermore, the system utilizes analytical redundancy to enhance mission robustness: in the event of an IRU hardware failure, the AOCS performs STT pseudo-differentiation to mathematically estimate angular velocity from the star trackers, guaranteeing continuous rate data [49].
- **Earth Sensor (ES):** Redundant configuration (ESH-1/ESE-1, ESH-2/ESE-2) utilized as a primary Earth-pointing reference during the Standard Attitude Control phase, operating in conjunction with the FSSs and IRU [27, 51]. Consistent with GOSAT's stringent robustness and survivability philosophy, the ES also serves as a critical FDIR backup: the AOCS is designed to rely on the ES to maintain attitude determination in the severe event of a complete failure across all STTs [49].
- **Fine Sun Sensor Assembly (FSSA):** Fully redundant configuration comprised of two processing electronics (FSSE-A, FSSE-B) and six optical heads (FSSH-1A, 2A, 3A and FSSH-1B, 2B, 3B) [27]. The FSSs are utilized as essential attitude sensors during the initial operations, specifically to establish and maintain the Standard Attitude Control Mode in conjunction with the Earth sensors and IRU, prior to the transition to the high-precision attitude mode [51].
- **Global Positioning System Receiver (GPSR):** Redundant configuration (GPSR-A, GPSR-B) utilized as a primary sensor during the High-Precision Attitude Control Mode, operating in conjunction with the STTs and IRU [51]. Crucially, the GPSR provides the on-board orbit determination values required by the AOCS to execute the autonomous orbit control function, which maintains the satellite's east-west orbit within strict ± 2.5 km margins [27, 49].

6.4 Attitude Control & Actuators

GOSAT controls its attitude using reaction wheels and magnetic torquers, while AOCS is also connected to the RCS to support the execution of control commands and ensure pointing requirements across all mission phases.

- **Reaction Wheel Assembly (RWA):** Four wheels (RWA-1 to RWA-4) in a redundant configuration serve as the primary actuators for 3-axis attitude control during all nominal and fine-pointing operations (e.g., Standard Attitude, High-Precision, and Moon Pointing modes) [27, 49]. A critical design feature, validated by Structural Thermal Model (STM) ground testing and flight data, is the management of mechanical micro-vibrations. To prevent RW-induced disturbances from affecting the IRU and the highly sensitive scientific payloads, the wheels operate within a specific rotational speed regime around 800 to 1000 rpm. In this region, on-orbit

measurements perfectly match ground predictions, confirming that the induced angular velocity noise is absolutely negligible [49]. Furthermore, the AOCS FDIR architecture ensures high survivability: in the severe event of multiple RW failures, the system is designed to autonomously fall back to the propulsion system (thrusters) for attitude control [49].

- **Magnetic Torquers (MTQs):** Configured with two electromagnetic units (MTQ-1, MTQ-2) driven by cross-strapped redundant electronics circuits (MTQE-A, MTQE-B) [27]. According to the AOCS design architecture, the MTQs are utilized to continuously unload the accumulated angular momentum from the RWs [49].
- **RCS Thrusters:** Dual-thrust propulsion system comprised of 20 N thrusters for large orbit control maneuvers (e.g., out-of-plane and major in-plane corrections) and 1 N thrusters for fine orbit maintenance [51]. The RCS is used during the initial critical operations for detumbling, immediately after launch vehicle separation [51]. Furthermore, operating as the ultimate FDIR backup, the thrusters are designed to substitute the RWs for 3-axis attitude control in the severe event of multiple wheel failures [49]. A highly optimized architectural feature of GOSAT is its thruster geometry: rather than relying on a conventional 6-thruster minimum configuration, the system achieves full 3-axis attitude control using only 4 thrusters. By arranging them in an inclined configuration to create a tetrahedral torque vector space, the AOCS can generate arbitrary 3-axis torques by activating at most three thrusters. Conversely, simultaneous firing of all four thrusters systematically cancels the torque components, producing pure X-axis translational thrust for orbit control [49].

7 AOCS Sizing

7.1 Inertia Approximation

To estimate the spacecraft inertia tensor, GOSAT was approximated as a rigid cuboid main body equipped with two rectangular solar arrays. The spacecraft bus dimensions were taken as $3.7 \times 1.8 \times 2.0$ m [3], while the solar array dimensions were estimated from the deployed spacecraft geometry reported in [52]. Each solar array was modelled as a thin rectangular plate of approximate dimensions 5.0×2.8 m and assumed to be symmetrically mounted along the spacecraft principal symmetry axis.

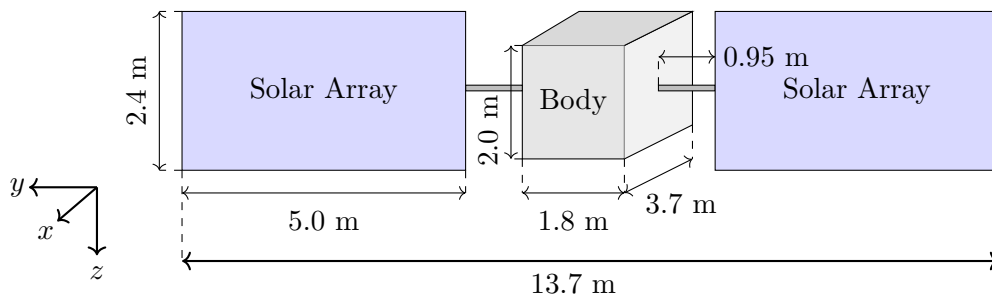


Figure 14: Simplified geometric approximation of the GOSAT spacecraft used for inertia estimation.

The solar array mass was estimated from the spacecraft power requirement of 3.8 kW [3] and a representative specific power of $80 \frac{W}{kg}$ for triple-junction GaAs solar cells used on GOSAT [Sharp_solar_cells]. Considering the capability to operate with a single solar panel [27], the total solar array mass was conservatively estimated as 100 kg, yielding an approximate body mass of 1650 kg.

Using standard analytical expressions for cuboids and thin plates, the resulting principal moments of inertia were obtained as: $I_x = 3161.42$ $I_y = 2497.71$ $I_z = 4428.46$ [$kg \cdot m^2$]

7.2 Disturbances

Various disturbance torques act on the satellite, including solar radiation pressure, gravity-gradient effects, atmospheric drag, and magnetic interaction with Earth's field. Their estimated magnitudes are summarized in Table 10.

Gravity Gradient			Solar Radiation Pressure					Air Drag				Mag. Field	
R	ΔI	θ	F_s	A	q	i	Δc_{SRP}	ρ	c_d	v	Δc_{AD}	D	B
[km]	[kg m ²]	[deg]	[W/m ²]	[m ²]	[-]	[deg]	[m]	[kg/m ³]	[-]	[km/s]	[m]	[A m ²]	[μT]
7037	1930.8	89	1367	29.6	0.3	0	0.1	$5.802 \cdot 10^{-14}$	2.5	7.53	0.1	5.72	45.7
$T_{GG} = 1.156 \cdot 10^{-4}$			$T_{SRP} = 1.753 \cdot 10^{-5}$					$T_{AD} = 1.216 \cdot 10^{-5}$				$T_{MAG} = 2.613 \cdot 10^{-4}$	

Table 10: Variables used in disturbance torque equations and disturbance torques summary [Nm]

To evaluate the gravity gradient torque, the angle θ between the minimum inertia axis and the spacecraft radial direction was needed and so a value of $\theta = 89^\circ$ was chosen to represent highly accurate nominal pointing while still accounting for slight inertia-axis misalignment, non-ideal mass distribution, or structural offsets. Also, Δc_{SRP} and Δc_{AD} are assumed to be 0.1 m from the geometrical configuration. ρ is the atmospheric density computed using a standard exponential model. The magnetic dipole can be estimated according to [53] considering the S/C dry-mass, $D = 3.5 \cdot 10^{-3} \cdot M_{dry} = 5.72 \text{ Am}^2$.

7.3 Sensors and Actuators selection

Since no detailed information is available in the literature regarding the exact sensors used by GOSAT, Table 11 presents a representative set of sensors consistent with the performance class required by the mission. IRU's stability is sufficient to satisfy requirements for inertial navigation during the Moon pointing mode. As for the actuators, the selected RW are those adopted on GOSAT-2, commissioned a few years later but characterized by similar physical and mission requirements, while the MTQ selection is later justified in subsection 7.5.

Sensors		
STT [54]	Accuracy	$< 3''$
ES [55]	Accuracy	$< 0.04^\circ$
FSS [56]	Accuracy	$< 0.3^\circ$
IRU [57]	Short term stability	$< 0.006^\circ/\text{h}$
Actuators		
Reaction Wheels [58]	$h_{max} = 30 \text{ [Nms]}$	$T_{max} = 0.4 \text{ [Nm]}$
Thrusters [17]	$T_{max} = 1 \text{ [N]}$	$L = 0.5 \text{ [m]}$
Magnetorquers [59]	$D = 140 \text{ [Am}^2\text{]}$	

Table 11: Selected sensors and actuators performance

7.4 RW Sizing

To verify that the selected RW are compatible with the mission requirements, both saturation and slew performance must be assessed. Considering the desaturation case, only constant disturbances are accounted for, since cyclic components cancel out over each orbit, with a safety factor of 2 adopted to account for the preliminary nature of the disturbance estimation. The angular momentum accumulated per orbit h_{orbit} is then evaluated by multiplying the resulting disturbance torque by the orbital period.

$$T_{dis} = 2(T_{gg} + T_{drag}) = 2.56 \cdot 10^{-4} \text{ Nm}, \quad h_{orbit} = T_{dis} \cdot T_{orb} = 1.50 \text{ Nms} \quad (10)$$

Given the reaction wheels' maximum momentum storage capacity of $h_{\max} = 30\sqrt{2}$ Nms, where the factor $\sqrt{2}$ derives from the effective momentum capacity of a four-wheel pyramidal configuration projected in the transverse plane, the number of orbits the spacecraft can complete before requiring a momentum dumping maneuver can be determined as follows:

$$N_{\text{orbits}} = \frac{h_{\max}}{h_{\text{orbit}}} = 28.26 \quad (11)$$

In nominal conditions, attitude maneuvers are performed using the reaction wheels. In particular, slew maneuvers are sized assuming a maximum angular rate of the S/C of $0.5^\circ/\text{s}$, which is used to verify that the selected wheels can provide the required torque and momentum without reaching saturation limits. Assuming a worst case scenario of a $\theta_{\text{slew}} = 180^\circ$ rotation and considering the maximum torque that the RWs can provide, the minimum slew time as well as the associated angular rate and stored momentum can be computed as:

$$t_{\text{slew}} = \sqrt{\frac{4\theta_{\text{slew}}I_{\max}}{T_{\max}}} = 372.99 \text{ s}, \quad \dot{\theta}_{\text{slew}} = \frac{T_{\max}}{I_{\max}} \frac{t_{\text{slew}}}{2} = 0.97^\circ/\text{s}, \quad h_{\text{slew}} = I_{\max} \cdot \dot{\theta}_{\text{slew}} = 74.6 \text{ Nms} \quad (12)$$

Since the resulting angular rate exceeds the maximum allowable value, it can be concluded that $\dot{\theta}_{\text{slew}}$ is the limiting factor for the maneuver, rather than the available torque. Assuming instead an angular rate of $0.3^\circ/\text{s}$, the corresponding maneuver duration, required torque, and accumulated angular momentum remain within the respective limits: $t_{\text{slew}} = 1200 \text{ s}$, $T_{\text{slew}} = 0.039 \text{ Nm}$, $h_{\text{slew}} = 23.19 \text{ Nms}$.

7.5 MTQ Sizing

In order to keep the RWs within the nominal operating range and avoid interference with TANSO measurements, the MTQ continuously desaturates the wheels by compensating for accumulated disturbances. Considering the sum of all constant torques and applying a safety factor of 5, introduced both to consider the lack of complete directional control and to account for the limited accuracy of the disturbance model, the required dipole moment of the MTQ can be computed:

$$D_{\text{mtq}} = \frac{5T_{\text{dis}}}{B_{\text{eq}}} = 55.94 \text{ Am}^2 \quad (13)$$

where the equatorial B has been considered, as it is the most critical condition to generate a torque. This result is consistent with the selection of the MTQ presented in Table 11. The component appears oversized due to the fact that all candidate options within the required range are custom-built solutions, whereas the present selection is based on the datasheet of a commercially available device.

7.6 Thrusters Sizing

While slew maneuvers are nominally performed using RWs, thrusters are used during Sun Acquisition mode (Safe mode), as they require lower power consumption. In this case, the maneuver is divided into three phases: acceleration, coasting, and braking, with $n = 2$ thrusters considered per maneuver. Assuming once again the worst-case scenario of $\theta_{\text{slew}} = \theta_{\text{acc}} + \theta_{\text{coast}} + \theta_{\text{brake}} = 180^\circ$, and a maximum angular rate of $\dot{\theta} = 0.5^\circ/\text{s}$ as stated in Table 9, since no wheel saturation limit has to be considered, the acceleration and coasting times can be computed as:

$$t_{\text{acc}} = t_{\text{brake}} = \frac{\dot{\theta}I_{\max}}{nFL} = 38.65 \text{ s}, \quad t_{\text{coast}} = \left(\theta_{\text{slew}} - \frac{nFL}{I_{\max}} t_{\text{acc}}^2 \right) \frac{1}{\dot{\theta}} = 321.35 \text{ s} \quad (14)$$

The total time and the total mass of propellant needed for a single Sun Pointing event is given by:

$$t_{\text{tot}} = t_{\text{acc}} + t_{\text{coast}} + t_{\text{brake}} = 398.65 \text{ s} \quad M_{\text{prop}} = \frac{n(t_{\text{acc}} + t_{\text{brake}})F}{I_{\text{sg0}}} = 0.0716 \text{ kg} \quad (15)$$

7.7 Positioning of the sensors and actuators

During nominal operations, the Sun follows a repetitive trajectory in the body-fixed coordinate system due to GOSAT's sun-synchronous orbit. The Sun path was calculated over one sun-synchronous orbit with a local time of descending node at 13:30 PM by tracking the Sun position in the body frame, while the Earth horizon envelope was determined assuming constant orbit altitude and a spherical Earth model, as shown in Figure 15. Taking this into account, both the FSSs and STTs can be positioned accordingly. The fields of view of the STTs are approximated from their mounting position on the S/C body [60] and the adopted sensor model [54]. The two STTs are oriented toward deep space while ensuring that their field of view avoids direct Sun exposure. In contrast, the FSSs are arranged to provide complete Sun coverage during nominal conditions as well as full coverage during maneuvers. Two ESs are mounted facing opposite directions to observe opposing edges of the Earth during nominal observations and in the surrounding region during attitude maneuvers. Finally, two GPSRs are installed on the face opposite to nadir, since GNSS satellites operate at significantly higher altitudes than GOSAT.

RCS components are placed on the (-)X-side of the satellite for routing convenience. The reaction wheel assembly, arranged in a pyramid configuration, is positioned centrally, while direct structural coupling with the payload is avoided in order to reduce the transmission of microvibrations. Two magnetic torquers are mounted along the edges of the spacecraft, with their dipole moments possibly aligned with the Y and Z-body axes. For a nadir-pointing S/C, the geomagnetic field is roughly aligned with the X-axis near the equator, so Y- and Z-axis torquers provide the greatest pitch and yaw control authority. Near the poles, the field aligns more closely with the Z-axis, increasing roll authority from the Y-axis torquer.

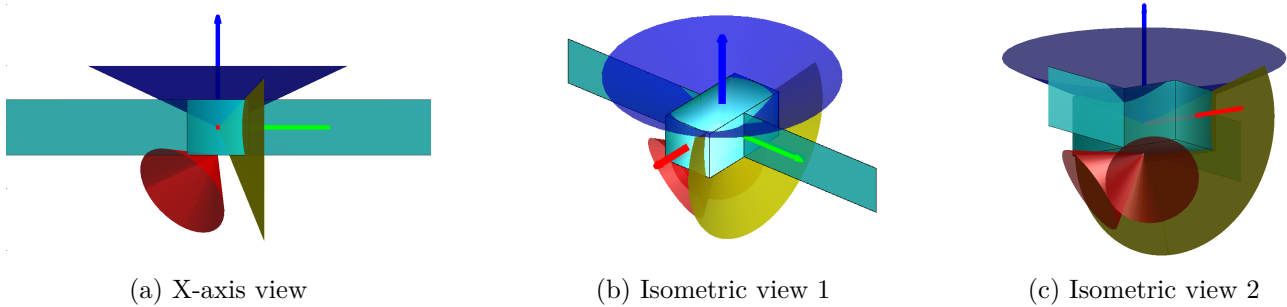


Figure 15: Yellow: sun path, Blue: Earth's horizon envelope, Red: star trackers fields of view.

7.8 Mass and power budget

The Mass and Power Budget of the AOCS subsystem is reported in Table 12.

	RW×4 [58]	MTQ×2 [59]	STT×2 [54]	ES×2 [55]	FSS×6 [56]	IRU [57]	GPSR×2 [61]
Mass [kg]	8.6	3.8	0.37	1.5	0.375	6.6	0.031
Power [W]	18 - 230	4.5	0.9	0.8	0.25	36	1.8

Table 12: Mass and Power budget (considering a single component)

8 Electrical Power System (EPS) Architecture

8.1 Overview

The Electrical Power System (EPS) of GOSAT employs a centralized regulated architecture, comprising a 50 V bus, solar arrays for power generation during sunlit orbital phases and Ni-Cd batteries for energy storage supply during eclipses. The design prioritizes high redundancy, ensuring partial operational continuity even in the event of a single solar panel failure through a symmetric configuration as can be seen in Figure 16.

Power control is dynamically adjusted based on the spacecraft’s operational mode and orbital position, along its SSO orbit. As detailed in [62], the operational modes and transition logic are supported by continuous telemetry monitoring of bus voltage, battery depth of discharge, and shunt currents to ensure system stability.

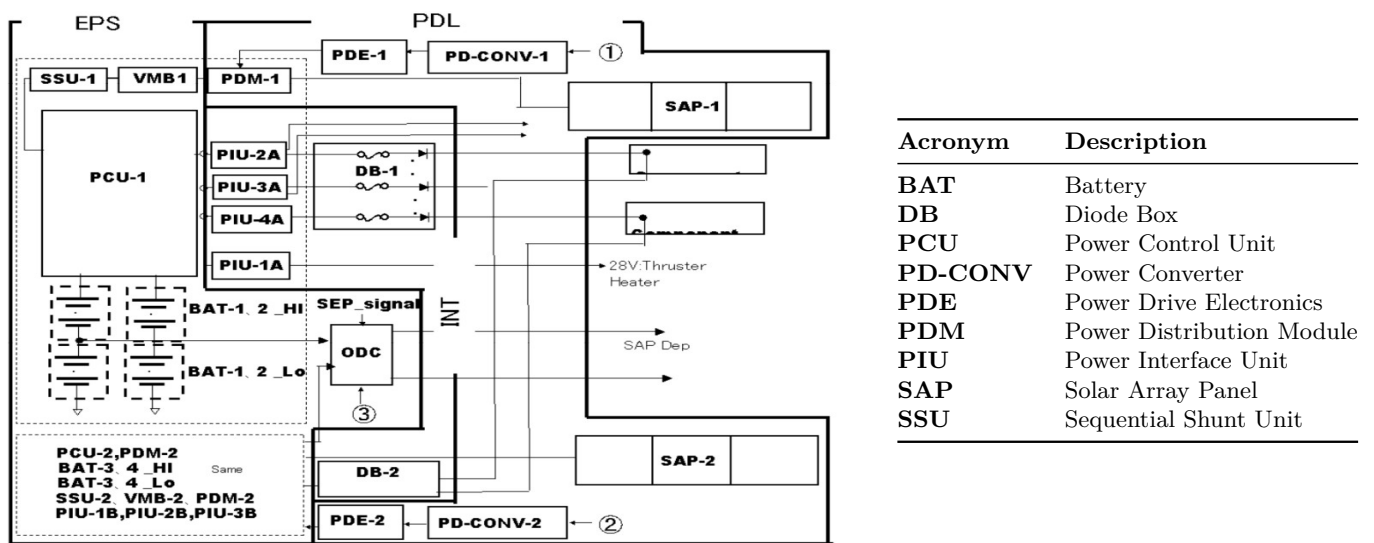


Figure 16: EPS Architecture [63]

8.2 Power demand & consumption

The power demand of all subsystems during the different modes is described below:

- **Payload (PL):** Regarding the PL, the main power consumption is associated with the TANSO instrument, whose nominal power requirements are 400 W for the FTS and 83 W for the CAI, during daytime, as stated in [64, 65]. In order to reconstruct the instrument’s nighttime power usage, since no public data is available, a nighttime to daylight power ratio of 0.63 is assumed, which is consistent with the GOSAT-2 mission, for which a more detailed breakdown is available [66]. Finally, when the instrument is not observing, the resulting power demand is even lower.
- **PS:** The PS is nominally inactive during instrument measurements and is only used to correct orbital parameters via the 20 N thrusters or to maintain Sun pointing in critical conditions using the 1 N thrusters, due to their lower power demand compared to the RWs.
- **AOCS:** Sensors and actuators are used as described in the AOCS chapter, and in particular steady-state power demand for the RWs has been considered for all modes except in the slew case, where maximum power consumption is assumed.
- **TMTC:** S-band telemetry and command communication is assumed to be active in all modes, while scientific data downlink is used only in observation mode. In particular, for power budget analysis, X-band communication is assumed to be active, as it represents the most demanding operational condition.

- **Thermal Control System (TCS):** Since in GOSAT's case the majority of the observing equipment is active only during daytime, it is reasonable to assume that the most demanding condition for the TCS occurs during the sunlight phase, due to the need to maintain instrument temperatures within their operational ranges. Reasonable values have been assumed by scaling the power reported in [62, Table 4.3-1] by the lower overall payload power of GOSAT, yielding values consistent with the TCS power fraction of approximately 10 % found in [9, Table 14-20].
- **EPS & On Board Data Handling (OBDH):** Due to the absence of literature values, a preliminary power budget for these subsystems has been allocated based on [9, Table 14-20], where fractions of 9 % and 12 % are assigned to the EPS and OBDH, respectively, by first estimating the total spacecraft power consumption considering that the payload accounts for approximately 46 %. A 50 % reduction in OBDH power consumption is assumed in Safe mode.

Subsystem	Hardware	Observation	Orbit control	Slew	Eclipse	Safe
PL	TANSO-FTS	400	150	150	250	150
	TANSO-CAI	83	30	30	52	30
PS	1 N Thrusters	[-]	[-]	[-]	[-]	31.2
	20 N Thrusters	[-]	65	[-]	[-]	[-]
AOCS	IRU	36	36	36	36	36
	FSS	[-]	[-]	[-]	[-]	0.75
	GPSR	1.8	1.8	[-]	1.8	[-]
	STT	0.9	0.9	[-]	0.9	[-]
	RW	140	140	460	140	[-]
	MTQ	9	9	[-]	9	[-]
TMTC	S-Band	26	26	26	26	26
	X-Band	98	[-]	[-]	[-]	[-]
TCS	[-]	120	120	120	75	75
EPS	[-]	95	95	95	95	95
OBDH	[-]	130	130	130	130	65
Total	[-]	1139.7	803.7	1047	815.7	508.95
Total + 20% margin	[-]	1367.64	964.44	1256.4	978.84	610.74

Table 13: Power budget [W]

8.3 Power generation, storage, and distribution

8.3.1 EPS Bus

The GOSAT EPS adopts a fully regulated dual-bus architecture [16] with redundant power distribution lines, as shown in the bottom left corner of Figure 16 with the use of the caption "Same". Electrical power generated by the solar array paddles is conditioned through dedicated Power Conditioning Units, which regulate the spacecraft bus voltage at 50 V and manage the charge and discharge of the Ni-Cd batteries. Voltage level is chosen considering the S/C's power demand [67] and standardized values provided in [62].

The EPS utilizes a Direct Energy Transfer (DET) configuration, as the solar arrays are not connected to any peak power tracking device. This choice is consistent with the mission objective of minimizing potential failure points as it offers a simpler and more robust design, compared to peak power tracking (PPT) architectures. In particular, the EPS is composed of two independent and symmetric power chains to improve fault tolerance and survivability. In case of failure of one bus line, spacecraft loads can still be powered through the remaining operational bus via the diode-isolated distribution network [63]. This redundant design also enables the EPS to reliably support critical mission phases, including the activation of pyrotechnic solar panel Hold Down & Release Mechanisms (HDRMs) and

the RCS operations [63]. Furthermore, control algorithm is designed in a way to preserve operation of the satellite under unforeseen circumstances by automatically shutting-down payload power supply during critical solar panel malfunctions [68].

8.3.2 Secondary Battery

The GOSAT EPS employs eight Ni-Cd battery units, arranged in four sets of 35 Ah each, to provide energy storage during eclipse operations. Splitting the batteries into smaller packs allows for more freedom in their placement in the S/C and simplifies the battery management system design. Ni-Cd technology was selected due to its extensive flight heritage, high reliability, and long calendar life of up to 15 years, making it particularly suitable for long-duration LEO missions. Compared to Ni-H₂ batteries, Ni-Cd offer higher volumetric energy density and higher operating temperatures, simplifying the thermal control architecture and reducing cooling requirements [69]. However, their performance is strongly temperature dependent, requiring operation within a narrow range typically between -5°C and $+10^{\circ}\text{C}$ to maintain high charge efficiency while limiting self-discharge. Furthermore, the maximum depth of discharge was limited to 20% [51] in order to maximize cycle life and ensure high long-term reliability. The main drawback of Ni-Cd cells is the memory effect, which requires periodic reconditioning through controlled discharge and recharge cycles to recover the nominal capacity [69].

8.3.3 Solar Cells

The GOSAT solar array utilises space-qualified InGaP/GaAs/Ge standard triple-junction solar cells produced by Sharp [70], selected for their high efficiency, reliability and extensive performance history. According to manufacturer datasheet [71], the cells achieve a Beginning Of Life (BOL) efficiency of 28.7% and maintain an End of Life (EOL) efficiency of 25.1% under AM0 conditions [72].

The selected solar cell configuration is based on the $A_{\text{cell}} = 27.6 \text{ cm}^2$ variant reported in the manufacturer datasheet [71], which provides a maximum voltage of $V_{\text{cell}} = 2.33 \text{ V}$ and a maximum power output of $P_{\text{cell}} = 1.081 \text{ W}$ under AM0 conditions at 25°C . The selected cell size was chosen as it is consistent with the overall dimensions of the solar array adopted in the spacecraft configuration.

8.3.4 Paddle Subsystem

The solar panels are connected to the satellite by paddles, rotating around the S/C's y-axis, that allow them to track the Sun. Since the satellite is Earth-pointing on an SSO whose orbital plane is not perpendicular to Sun's direction, fixed panels would not be able to optimize power generation, as they cannot maintain the correct orientation.

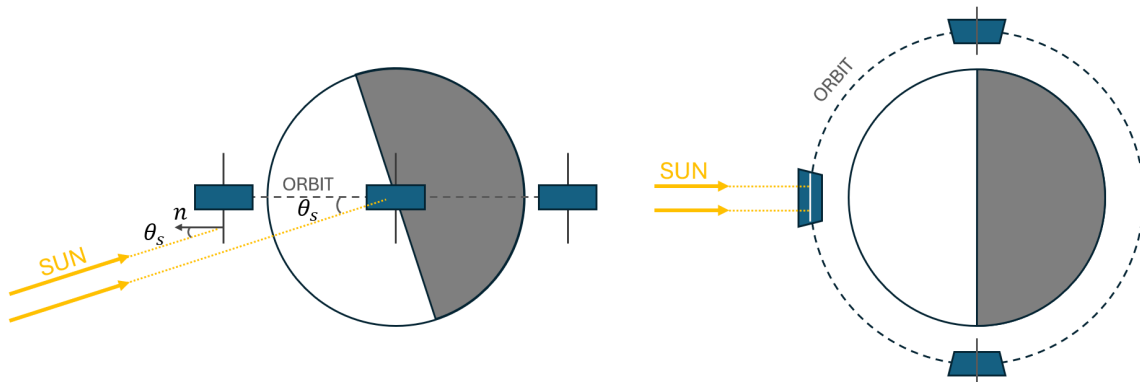


Figure 17: Sun Aspect Angle (SAA) scheme a) Top view b) Side view

These paddles also feature a deployment system that keeps the panels folded during launch and extends them once in orbit. For a satellite in LEO, the mean radiation flux is assumed to be the same as that

reaching the Earth, namely varying between 1321 - 1422 W/m² throughout the year. Indeed, with respect to the Sun-Earth distance (1 AU), the orbital radius of the spacecraft is negligible. Given that the SSO orbit has an Local Time of Descending Node (LTDN) of 13:30, the angle between the Sun's direction and the orbital plane, the Sun Aspect Angle (SAA), is constant by definition of an SSO, and can be computed as $\theta_s = \frac{360^\circ}{24 \text{ h}} \cdot 13.5 \text{ h} - 180^\circ = 22.5^\circ$. Based on the calculated Sun visibility in the S/C's reference frame, it can be derived that solar arrays has to rotate by approximately 229° every orbit, powered by a redundant set of electric motors [73].

9 Electrical Power System (EPS) Sizing

9.1 Solar Panel Sizing

The solar array sizing was conducted based on two primary criteria: the mission power requirement and GOSAT's reliability demand. The first defines the power needed to support nominal operations, while the second accounts for redundancy constraints, which ultimately drives the sizing of the EPS.

9.1.1 Sizing based on power budget

The first step in sizing the solar array is defining the worst-case scenario. This involves identifying the maximum eclipse duration ($T_e = 0.5792 \text{ h}$), the corresponding daylight period ($T_s = 1.0562 \text{ h}$), and the peak power requirements during both eclipse (P_e) and daylight¹⁰ (P_s). Based on these parameters, the maximum power required from the solar array (P_{sa}) is calculated as follows:

$$P_{sa} = \frac{P_e T_e}{X_e T_s} + \frac{P_s}{X_s} = 2435 \text{ W} \quad (16)$$

where X_e and X_s represent the efficiency factors during eclipse and daylight, respectively, and they depend on the adopted power control architecture. Specifically, GOSAT employs a DET architecture, as illustrated in subsubsection 8.3.1, resulting in $X_e = 0.65$ and $X_s = 0.85$ [9].

The specific power generated by the solar panels at BOL is computed as:

$$P_{BOL} = P_0 \cdot \varepsilon \cdot I_d \cdot \cos(\theta_s) = 245.19 \frac{\text{W}}{\text{m}^2} \quad (17)$$

Here, the worst-case solar constant for the spacecraft-Sun distance is $P_0 = 1321 \text{ W/m}^2$. The solar cell efficiency, sourced from the catalog, is $\varepsilon = 0.287$ [71]. The inherent degradation factor, which accounts for initial cell losses, is set to $I_d = 0.7$, representing the median of the validity range for multijunction cells, which is appropriate given the use of a triple-junction architecture.

The EOL specific power is obtained by accounting for the yearly degradation over the mission lifetime:

$$P_{EOL} = P_{BOL} \cdot (1 - dpy)^{N_{years}} = 210.55 \frac{\text{W}}{\text{m}^2} \quad (18)$$

where the degradation per year is assumed to be $dpy = 0.03$ (typical multi-junction solar cell value) and the mission lifetime is $N_{years} = 5$.

Using the maximum required power and the EOL specific power, the minimum required solar array area (A_{sa}) and the corresponding number of cells (N_{cells}) are determined:

$$A_{sa} = \frac{P_{sa}}{P_{EOL}} = 11.56 \text{ m}^2 \quad N_{cells} = \left\lceil \frac{A_{sa}}{A_{cell}} \right\rceil = 4190 \quad (19)$$

where $A_{cell} = 27.6 \text{ cm}^2$ [71].

This preliminary sizing is then refined to establish the exact number of cells and their series-parallel

¹⁰observation mode

configuration. The number of cells connected in series is dictated by the bus voltage level. Given the 50 V regulated bus, the required series connections and parallel strings are:

$$N_{series} = \left\lceil \frac{V_{bus}}{V_{cell}} \right\rceil = 22 \quad N_{parallel} = \left\lceil \frac{N_{cells}}{N_{series}} \right\rceil + 1 = 192 \quad (20)$$

Consequently, the final number of cells and the effective solar array area are:

$$N_{cells,real} = N_{parallel} \cdot N_{series} = 4224 \quad A_{sa,real} = A_{cell} \cdot N_{cells,real} = 11.66 \text{ m}^2 \quad (21)$$

The actual power generation capability of this sized solar array at BoL and EoL, assuming perpendicularity to the Sun, is evaluated as follows:

$$P_{sa,BOL} = A_{sa,real} \cdot P_0 \cdot \varepsilon \cdot I_d = 3094 \text{ W} \quad P_{sa,EOL} = P_{sa,BOL} \cdot (1 - dpy)^{N_{years}} = 2657 \text{ W} \quad (22)$$

9.1.2 Sizing based on mission requirements

A secondary sizing approach addresses the mission's redundancy requirements. Specifically, the system must support a degraded observation mode to partially continue the mission even in the event of a total loss of one solar array wing [27]. To satisfy this requirement, the remaining solar array must generate over 2 kW at EOL with the condition that the sunlight is normal to the paddle surface [16]. Therefore, taking 2 kW as reference value, this requirement based sizing begins with a $P_{sa,single,EOL} = 2000 \cdot \cos(\theta_s) = 1848 \text{ W}$.

The corresponding BoL power requirement is computed by factoring in the expected degradation over the mission duration:

$$P_{sa,single,BOL} = \frac{P_{sa,single,EOL}}{(1 - dpy)^{N_{years}}} = 2152 \text{ W} \quad (23)$$

The necessary area and number of cells for a single wing are then calculated as:

$$A_{sa,single,real} = \frac{P_{sa,single,BOL}}{P_0 \cdot \varepsilon \cdot I_d} = 8.108 \text{ m}^2 \quad N_{cells,single,real} = \left\lceil \frac{A_{sa,single,real}}{A_{cell}} \right\rceil = 2938 \quad (24)$$

From which the total number of cells and total area for both symmetric solar array wings can be recovered:

$$N_{cells,total} = 2 \cdot N_{cells,single,real} = 5876 \quad A_{sa,total} = 2 \cdot A_{sa,single,real} = 16.22 \text{ m}^2 \quad (25)$$

Finally, a preliminary mass estimation is performed using the specific power coefficient typical for triple-junction solar array cells:

$$SP = 80 \frac{\text{W}}{\text{kg}} \quad M = \frac{2 \cdot P_{sa,single,BOL}}{SP} = 53.79 \text{ kg} \quad (26)$$

9.1.3 Comparison with real values

The real solar array characteristics were estimated from publicly available images and operational data of the GOSAT [74]. From the visible cell layout, a total of 6384 cells was estimated, corresponding to an approximate solar array surface area of 17.62 m² assuming a cell area of 27.6 cm² [71]. The initial sizing, based solely on the power budget, underestimates the actual solar array capabilities because the physical system was ultimately designed to satisfy strict redundancy criteria. The secondary sizing approach, which explicitly incorporates these requirements, confirms this rationale by producing values that closely match the actual spacecraft specifications as seen in Table 14.

	Power Budget Sizing	Requirement Sizing	Literature Values [2]
Power (EOL) [W]	2657	3696	3770
Mass [kg]	38.68	53.79	[-]

Table 14: Power and mass budget

9.2 Battery Sizing

The spacecraft is equipped with batteries that provide the power required by the subsystems during the eclipse phase. The battery is sized for a regulated bus voltage of $V_{\text{bus}} = 50 \text{ V}$, using individual cells with nominal voltage $V_{\text{cell}} = 1.4 \text{ V}$ and capacity $C_{\text{cell}} = 9 \text{ Ah}$ [75]. Considering the power requested in the eclipse phase computed in Table 13, an eclipse duration of $T_e = 0.5792 \text{ h}$ and a Depth of Discharge (DoD) of 20 % [51], it's possible to compute the minimum energy required by a single battery as:

$$E_{\text{batt}} = \frac{P_e \cdot T_e}{X_e \cdot N \cdot \eta \cdot \text{DoD}} = 1362.85 \text{ Wh} \quad (27)$$

where $N = 4$ is the number of batteries [27], and typical values of 0.65 and 0.8 have been assumed for X_e for a Direct Energy Transfer bus and η (EOL efficiency) respectively.

The next step is to define the electrical configuration of the battery. Cells are connected in series to reach the required bus voltage of 50 V. Since each selected cell provides 1.4 V, the number of cells required in series and the actual voltage of the battery can be computed as:

$$N_{\text{series}} = \left\lceil \frac{V_{\text{bus}}}{V_{\text{cell}}} \right\rceil = 36 \quad V_{\text{string}} = N_{\text{series}} \cdot V_{\text{cell}} = 50.4 \text{ V} \quad (28)$$

After fixing the series configuration, the capacity and the energy storable in one string are evaluated. A packing efficiency loss factor $\mu = 0.8$ is applied to account for practical losses and non-idealities in the battery assembly:

$$C_{\text{string}} = \mu \cdot C_{\text{cell}} = 7.2 \text{ Ah} \quad E_{\text{string}} = C_{\text{string}} \cdot (V_{\text{cell}} \cdot N_{\text{series}}) = 362.88 \text{ Wh} \quad (29)$$

Since one string alone is not sufficient to satisfy the required eclipse energy, multiple strings are connected in parallel, adding one additional string as margin. Finally, the total number of cells in a battery can be calculated:

$$N_{\text{parallel}} = \left\lceil \frac{E_{\text{batt}}}{E_{\text{string}}} \right\rceil + 1 = 5 \quad N_{\text{cells}} = N_{\text{parallel}} \cdot N_{\text{series}} = 180 \quad (30)$$

The actual energy and the electrical capacity of a single battery are then:

$$E_{\text{batt,real}} = N_{\text{parallel}} \cdot E_{\text{string}} = 1814.4 \text{ Wh} \quad C_{\text{batt}} = N_{\text{parallel}} \cdot \mu \cdot C_{\text{cell}} = 36 \text{ Ah} \quad (31)$$

The total mass is obtained, assuming an energy density of $\rho = 30 \text{ Wh/kg}$, typical for Ni-Cd batteries:

$$M_{\text{batt}} = N \cdot \frac{E_{\text{batt,real}}}{\rho} = 241.92 \text{ kg} \quad (32)$$

The mass obtained corresponds to less than 13.82% of the total mass, which is consistent with the estimates of the mass of the subsystems provided in [9, Table 14.18, p. 422]. Assuming single cell dimensions $53 \times 22 \times 94 \text{ mm}$ [75], the corresponding volume of all Ni-Cd cells is $V = 0.079 \text{ m}^3$, which corresponds to a cube of side length $l_{\text{side}} = 429 \text{ mm}$. It supports the benefit coming from splitting the battery volume into smaller sub-packs.

10 Thermal Control System (TCS) Architecture

The GOSAT's Thermal Control System (TCS) objective is to ensure that all spacecraft components remain within their allowable temperature ranges through a combination of passive and active thermal control. The main architecture relies on heat pipes, Multi Layer Insulation (MLI), and Optical Solar Reflectors (OSR), while active regulation is achieved via thermostatically controlled heaters. The payload mainly requires active cooling due to its optical nature.

10.1 Adopted Materials & Components

The spacecraft thermal insulation is based on MLI, which is assumed to be most likely composed of aluminized Kapton layers. MLI reduces radiative heat exchange by combining multiple low emissivity layers separated by low conductivity spacers, making it particularly effective in the space vacuum environment. Alternatively, local solutions found on GOSAT include additional MLI coatings with a copper like appearance as well as silver coating surrounding critical components such as tanks [19]. All antennas are featuring a white, RF-transparent reflective paint.

OSR are considered as part of the passive thermal control strategy and provide high solar reflectance and high infrared emissivity, allowing efficient rejection of internally dissipated heat while minimizing absorbed solar radiation. Typically they consist of a top layer made out of quartz glass with a reflecting metal layer beneath [76]. OSRs are commonly bonded onto radiator substrates and are among the most effective surface finishes for spacecraft radiators due to their low solar absorptivity and high infrared emissivity [76]. These surfaces are located on all visible faces of GOSAT.

10.2 Components temperature range and heat dissipation

The power dissipated as heat by the components corresponds to all or part of the power they receive to operate. Regarding temperature ranges, some components have an operating temperature range that differs from the survival temperature range. In such cases, both ranges are specified in Table 15.

OBDH & AOCS

Regarding purely electronic components, there is no mechanical work; 100% of the power consumed is dissipated as heat, that is, $P_{diss} = P_{elec}$. For RW, in steady-state, the power consumed is entirely converted into heat through internal mechanical friction. The value reported in Table 15 is the steady-state power.

TMTC

The X-band amplifier has an average efficiency of 34%. The power dissipated as heat is therefore $P_{diss} = (1 - \eta_{amp})P_{el} = 0.66P_{el}$, meaning that 66% of the consumed power is dissipated as heat. As for the S-band amplifier, since the average efficiency is not provided, a value of 40% is assumed, leading to $P_{diss} = 0.60P_{el}$.

Propulsion System

The thermal power dissipated by the thrusters and valves can be neglected in the global thermal analysis. Once in orbit, these components are only operated for very brief periods.

Payload: TANSO-FTS

The main sensing elements of the FTS are mounted on a thermally controlled bench maintained at 23 °C and constrained within a 20-26 °C range, with temperature variations limited to approximately 0.1K over the orbital calibration period. On the other hand components within the instrument with stricter thermal requirements employ dedicated active systems: a calibration diode laser is maintained at 25 ± 0.001 °C through a thermoelectric controller due to its wavelength sensitivity to temperature, while the four InGaAs detectors are cooled to -40 °C using thermoelectric coolers. The TIR detection

chain adopts an even lower operating temperature, with the HgCdTe detector being cooled to 70K by a pulse-tube cooler. [77]

Payload: TANSO-CAI

The CAI assembly is maintained at 23 ± 3 °C, the same temperature as the FTS, while its detectors adopt different thermal management strategies depending on their sensitivity. Silicon-based detectors operate without dedicated temperature control, relying on the stable thermal environment of the instrument, whereas the temperature sensitive InGaAs detector is actively regulated at 0 ± 0.5 °C using a thermoelectric cooler.

Battery

Ni-Cd batteries shall operate within very strict temperature ranges [0,-10 °C] to maximize their lifespan. The most common battery thermal designs use radiators and thermostatically controlled heaters to satisfy those strict requirements. Often fixed-conductance heat pipes are introduced to isothermalize the batteries and couple them to remote radiators [78]. As a result of those strict thermal requirements in overall system sizing usually less strict values are used [78, chap. 19, p. 717]. Moreover, although the battery does not consume electrical power to operate, it generates heat through Joule heating. However, as the charging current is low, this effect is neglected here.

Subsystem	Components	Operating T [°C]	Survival T [°C]	Dissipated Heat [W]
PL	TANSO-FTS	See TANSO-FTS	/	[250, 400]
	TANSO-CAI	See TANSO-CAI	/	[52, 83]
PS	1 N Thrusters	[+5, +90]	[-10, +95]	/
	20 N Thrusters [25]	[-10, +90]	/	/
	Valves (1N)[23]	[+4, +121]	/	/
	Valves (20N)[23]	[+4, +80]	/	/
	Tanks [79]	[+15, +50]	/	/
AOCS	IRU [57]	[-10, +50]	/	36
	FSS [56]	[-50, +85]	/	/
	ES [55]	[-30, +60]	/	/
	GPSR [61]	[-40, +85]	[-55, +95]	1.8
	STT [54]	[-40, +40]	[-40, +40]	0.9
	RW [58]	[-5, +60]	[-15, +60]	4×35
	MTQ [59]	[-40, +85]	/	2×4.5
TMTC	S-SSPA [9]	[-100, +100]	/	15.6
	X-SSPA [9]	[-100, +100]	/	[0, 66.7]
OBDH	FPGA [80] + other	[-55, +125]	[-65, +150]	130
EPS	Battery [78]	[0, +25]	[-10, +40]	/
	Solar Panels [9]	[-150, +110]	/	/
	Paddles & Electronics	/	/	95

Table 15: Subsystem Components and Specifications¹¹

Some components, such as the payload or the valves, have their own temperature regulation systems and should not be considered when determining the overall temperature limits of the satellite. The most restrictive maximum temperature for some components is 40°C; adding a 15°C uncertainty margin gives the maximum satellite temperature, $T_{sc,max} = 25^\circ\text{C} = 298.25$ K. Regarding the minimum reachable temperature, and excluding self-regulated components, the minimum allowable temperature is -10°C; adding the 15°C margin gives the minimum satellite temperature, $T_{sc,min} = 5^\circ\text{C} = 278.25$ K.

¹¹For the heat generation, parentheses include the dissipated power in the observation mode and eclipse mode (hot and cold case)

10.3 External Environment and Conditions

In LEO, main external heat sources acting on GOSAT are direct solar radiation, Earth's albedo, and Infrared (IR) radiation emitted by Earth. Direct solar radiation is the dominant contribution and is modeled using the solar constant $q_{sun} = 1367 \text{ W/m}^2$, while Earth albedo q_{albedo} accounts for the fraction of sunlight reflected by Earth back toward the spacecraft. They are clearly time dependent heat sources active for approximately one hour in each orbit.

IR radiation emitted by Earth q_{IR} results from the thermal re-emission of absorbed solar energy by Earth and constitutes an approximately constant thermal load on S/C surfaces through both shaded and sunlit part of orbit. Other environmental heating effects, such as free molecular heating and charged particle heating, are negligible at GOSAT's orbital altitude.

Since GOSAT operates in a SSO, pointing nadir in nominal mode, these environmental fluxes through each S/C surface vary repeatedly.

11 Thermal Control System (TCS) Sizing

The objective of this section is to size the satellite's thermal control subsystem to ensure that its temperature range remains within allowable limits. Firstly, the extreme cases have to be identified. The cold case occurs when the satellite receives and generates the least amount of energy; this happens during Earth's eclipse ($Q_{int,cold} = 730 \text{ W}$) during which the satellite is only exposed to Earth's infrared (IR) radiation and its own internal power dissipation. Conversely, the hot case occurs when the satellite receives and generates the highest amount of energy; this happens when it is fully exposed to the sun in observation mode ($Q_{int,hot} = 1174 \text{ W}$, computed with a 20% margin). The main constant parameters used in the sizing are presented in Table 16 and Table 17.

Earth radius	Albedo	Earth / Space Temp.	Emiss. IR
R_E [m]	a_E	T_E/T_{SP} [K]	ε_E
$6371 \cdot 10^3$	0.4	303.05 / 3	0.8

Table 16: Earth physical parameters

Material / Part	α	ε
Solar cell [81]	0.88	0.85
Radiator [78, Chapter 6, p.207]	0.15	0.85
Aluminized Kapton (MLI)[82]	0.14	0.05
Carbon fibre [83]	0.8	0.9

Table 17: Optical properties

11.1 Mononodal analysis

A mononodal thermal analysis is performed by modeling the S/C as an equivalent sphere with a similar surface area and homogeneous properties. The two worst case scenarios described previously are considered for this sizing.

First, a rough approximation is made, estimating that 80% of its surface is covered with MLI, while the remaining 20% consists of uncoated Carbon-Fiber-Reinforced Plastic Honeycombs. From this assumption, the global properties α_{sc} and ε_{sc} of the equivalent sphere are calculated.

As stated previously, the satellite is subjected to different types of heat fluxes. The thermal power received in each case is presented in Table 18, where Q_{int} is the internal power dissipation estimated in subsection 10.3. The thermal equilibrium equation yields a satellite temperature of $T_{sc,hot} = 398 \text{ K}$ during the hot case, which is well above the maximum allowable value.

	Q_{int} [W]	Q_{sun} [W]	Q_{albedo} [W]	Q_{IR} [W]
Hot Case	1174	3284	4307	2437
Cold Case	730	0	0	2437

Table 18: Thermal power

Adding radiators allows the system to dissipate heat more efficiently and thus lower this maximum temperature. Assuming that these radiators cover the 20% of the satellite's surface not protected by the MLI, new global α_{sc} and ε_{sc} constants are calculated. A new thermal equilibrium is then used to compute the additional radiator surface area required to remain below the maximum allowable temperature, which is increased by 5 K (reducing the 15 K margin to 10 K). The energy balance can then be written as :

$$Q_{sun} + Q_{albedo} + Q_{IR} + Q_{int} = Q_{emitted} + A_{rad}^{min} \sigma \varepsilon_{rad} ((T_{sc}^{max})^4 - T_{SP}^4) \quad (33)$$

where $\sigma = 5.67 \cdot 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$ is the Stefan-Boltzmann constant. The radiator area needed is $A_{rad}^{min} = 10.1 \text{ m}^2$.

Finally, this new configuration with the added radiators is used to determine the heater power required to guarantee that the satellite's temperature will not drop below the minimum required limit during the cold case. In this case, the heat balance is written as :

$$Q_{IR} + Q_{int} + Q_{heaters} = Q_{emitted}(T_{sc}^{min}) + Q_{rad}(T_{sc}^{min}) \quad (34)$$

The power required for the heaters, included 25% margin, is $Q_{heaters} = 3007 \text{ W}$. This value is significantly higher than the total consumption of the S/C during observation mode, then it is not acceptable.

Since GOSAT operates in LEO, its thermal environment changes cyclically and significantly between sunlit and shaded part of the orbit. As a result, a simulation of the S/C as a lumped mass of spherical shape, under a steady state condition in both coldest and hottest case, leads to unreasonable results drastically oversizing heaters and radiators. The preliminary TCS simulation shall be performed in a transient state, along the satellite's orbit. Therefore, sizing of the thermal control subsystems shall result in oscillating temperatures within the appropriate limits.

11.2 Multinodal transient analysis

In this section, a 3 nodes approach is adopted, featuring internal body, external surface, and solar array nodes, presented in Table 19. Specific heat capacities C_p of outer surface and inner body are approximately corresponding to aluminum alloy. C_p of solar array is assumed as a value between high performance martian rover solar array ($774 \frac{\text{J}}{\text{kgK}}$) and aluminum [84].

Node	Mass [kg]	C_p [J/kgK]	Position
1	1350	900	Main internal body
2	200	900	6 external (cuboid) body surfaces, calculated independently
3	200	800	Solar array (sunlit and shaded part treated separately)

Table 19: Thermal node definitions

Satellite body is modeled as a nadir pointing cuboid with solar panels constantly tracking sun along the full SSO. At each simulation step the angle between each panel normal and sun vector γ is tracked. The angle spanned between S/C position, Earth, and Sun vector is represented by ξ .

11.2.1 Heat fluxes

Solar radiation is calculated using the average value of the solar radiation flux during the year $q_{sun} = 1367.5 \frac{\text{W}}{\text{m}^2}$. Then, the solar radiation flux through each surface i is computed as $q_{sun-i} = q_{sun} \cos \gamma_i$.

Albedo and **IR** fluxes' form factors are calculated for surfaces facing nadir and perpendicular to nadir using the following equations [85]:

$$FF_{\text{nadir}} = \left(\frac{R_E}{r_{\text{norm}}}\right)^2 \quad FF_{\perp} = \frac{1}{2\pi} \left[\pi - 2 \arcsin \left(\sqrt{1 - \left(\frac{R_E}{r_{\text{norm}}}\right)^2} \right) - \sin \left(2 \arcsin \left(\sqrt{1 - \left(\frac{R_E}{r_{\text{norm}}}\right)^2} \right) \right) \right] \quad (35)$$

Since the solar array is neither aligned nor perpendicular with respect to nadir, a linear approximation between FF_{nadir} and $FF_{\perp}$ of its form factor is performed.

Then, the albedo flux is calculated: $Q_{\text{albedo}-i} = Q_{\text{sun}} a_E FF_i \cos \xi_i$; and IR flux is computed:

$$Q_{\text{ir}-i} = \varepsilon_E \sigma T_E^4 FF_i.$$

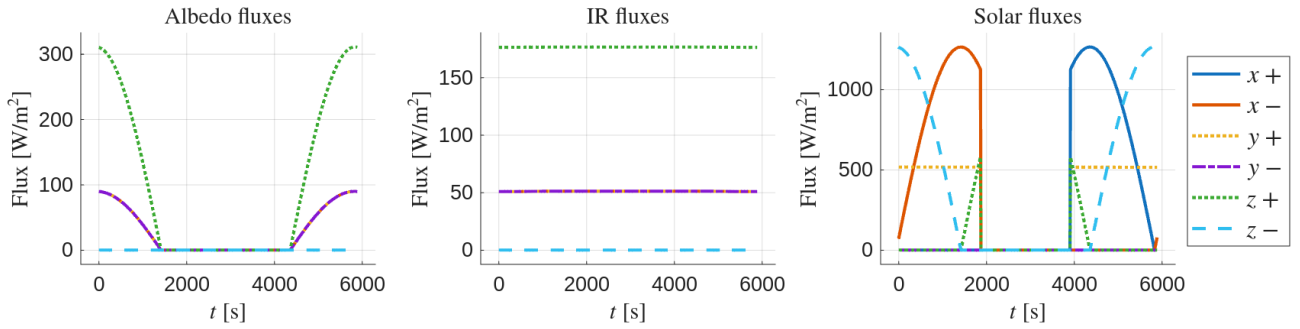


Figure 18: Fluxes through 6 main body surfaces over one orbit. Results validated with [85].

11.2.2 Absorbed heat power

As a preliminary analysis, a uniform distribution of radiators and thermal insulation is assumed to meet the temperature requirements and ensure stable oscillatory thermal behavior. As a result, three pairs of emissivity and absorptivity coefficients are calculated: one for the spacecraft outer body surface, one for the sun-facing side of the solar array, one for the non-sun-facing side of the solar array.

$$\alpha = a_{\text{RAD}} \alpha_{\text{RAD}} + a_{\text{coat}} \alpha_{\text{coat}} + a_{\text{SA}} \alpha_{\text{SA}}, \quad (36)$$

$$\varepsilon = a_{\text{RAD}} \varepsilon_{\text{RAD}} + a_{\text{coat}} \varepsilon_{\text{coat}} + a_{\text{SA}} \varepsilon_{\text{SA}} \quad (37)$$

Face	a_{RAD}	a_{coat}	a_{SA}
Body	0.21	0.79	0
SA +	0	0	1
SA -	0.50	0.50	0

Table 20: Surface fractions.

where a are the fractions of the surface occupied by the radiators, coating or solar cells, summarized in Table 20. Based on the selected materials and their distributions, the external heating power input to main body of the S/C is presented in Figure 19, alongside internal power production depending on the operation mode, which already accounts for estimated power of TCS [9].

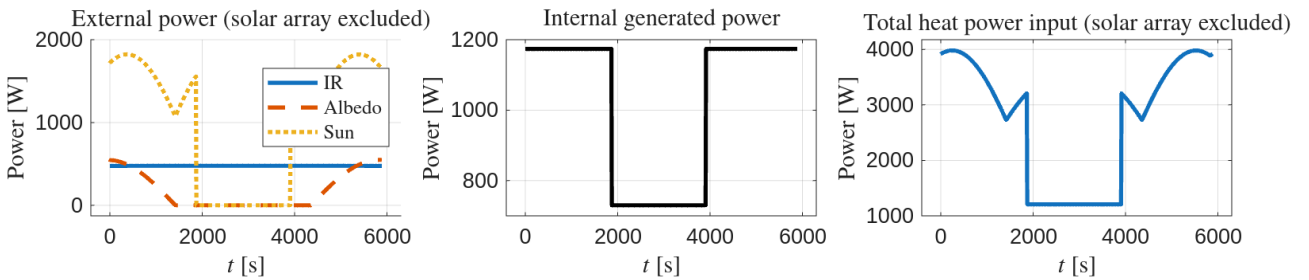


Figure 19: Heat power contributions.

A model of a series heat transfer is assumed, where solar array is connected to external body surface, which is then attached to the inner main body. First connection is assumed to be highly insulating with thermal conductance $G_{sa \leftrightarrow out} = 2[W/K]$. Second assumes conductance $G_{out \leftrightarrow in} = 143[W/K]$, which is achievable using thermal straps or with multiple bolted joint connections [78, Table 8.4].

11.2.3 Results analysis

As seen in Figure 20, temperature oscillations are stable and are bounded within nominal operation limits of all components. However, they solely represents bulk mass property, therefore local temperature control still needs to be assessed in the latter part of the development process. Moreover, it can be observed that the outer part of the body surface has a lower temperature than the inner body node, representing an outward overall heat flow in the spacecraft.

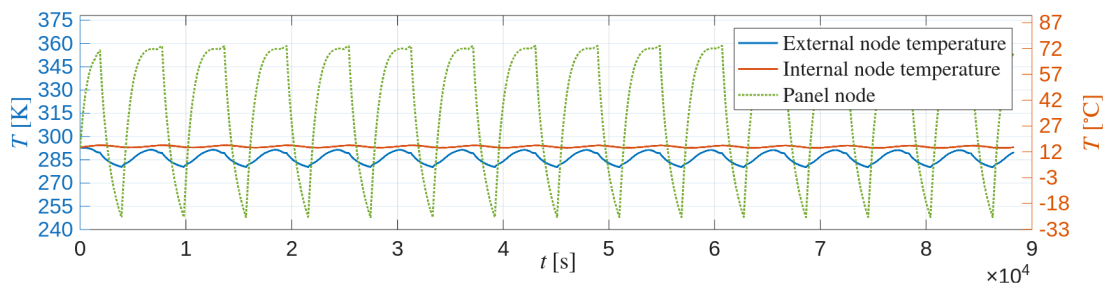


Figure 20: Temperature on nodes.

In the current simulation, the surface optical properties are constant over all main body surfaces, Figure 21 presents external normalized heat absorbed by each side of the S/C. Based on this result, a refinement of material distribution can be proposed, placing radiators on the surfaces with minimal heat values $\{x^+, x^-, y^-\}$. In reality however, radiator and heater placement is significantly influenced by specific components local temperature requirements and mounting possibilities. On top of that, even though surfaces x^+ and x^- have identical averaged heat flux values, they receive solar radiation power in different parts of orbit. Face pointing x^+ in the first, and x^- in the second half of the sunlit orbit section.

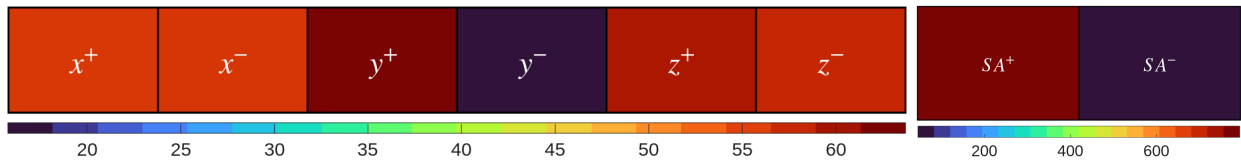


Figure 21: Average normalized external power to main body and solar array surfaces $[W/m^2]$.

11.3 Data & Mass & Power Budget

An estimate of the power and mass budget has been computed considering the preliminary sizing presented in [9, Table 14-18 and 14-20], due to the lack of detailed information about the TCS components mounted on GOSAT. In particular, the TCS accounts for approximately 2% of the S/C's dry mass, corresponding to 32.7 kg, and generally requires around 10% of the total power consumption, leading to 120 W. The TCS data budget can be estimated considering that a medium-sized satellite such as GOSAT typically employs approximately 250 temperature sensors [86], resulting in a data rate of about 4000 bps assuming a 1 Hz update frequency for housekeeping.

12 Structure and Configuration Subsystems

12.1 Introduction

The structural subsystem of GOSAT is designed to accommodate the Earth-observing optical payloads in a strict nadir-pointing configuration, protecting the delicate equipment from severe H-IIA launch loads while minimizing mass and thermal deformations in orbit. It utilizes a Carbon Fiber Reinforced Polymer (CFRP) primary cylinder and honeycomb sandwich panels, validated through a Proto-Flight Model (PFM) testing approach.

12.2 Spacecraft architecture

12.2.1 Modules configuration

The overall dimensions of the S/C are approximately 3.6m in height, 1.8m in width, and 2.0m in depth. It is divided into three physically separable modules to facilitate satellite assembly: the mission module, the bus module, and the deployable extensions.

Bus module: This module supports bus equipment. The central CFRP cylinder supports two 550mm tanks of the propulsion system (RCS) internally.

Payload module: This module houses the observation sensors (GOS, CS), TEDA, and the monitor camera (CAM).

Extension and propulsion modules: The satellite includes a PDL (Solar Array Paddle) system comprising three folding panels per wing (extending about 6m in the deployed state), a deployment mechanism, and a paddle drive mechanism.

12.2.2 Structure and materials

A CFRP monocoque cylinder is the primary structure of GOSAT and acts as the main load path between the spacecraft components and the launch vehicle. The main disadvantage of this configuration is its limited buckling strength; however, it is offset by a relatively low mass [87, Table 15.8].

The secondary structure is composed of CFRP sandwich panels connected primarily via angle brackets. These panels surround the primary structure on all sides and are attached to it by four rib panels as well as two circumferential rings at the top and bottom of the primary cylinder. The secondary structure also extends upward, supporting the payload, equipment, as well as the solar arrays. The panels consist of a honeycomb sandwich structure with CFRP skins, which reduces mass and minimizes thermal deformation, a critical aspect for ensuring precise instrument pointing. An aluminum honeycomb core provides high stiffness at low mass while also facilitating heat transfer to thermally controlled surfaces such as radiators.

The tertiary structures consist of box-shaped enclosures mounted on the secondary structure. They define distinct compartments for groups of components and also enable more controlled thermal management.

12.2.3 Internal and External layout

Satisfying payload functionality is the primary driver of the structural configuration of the S/C. In the case of GOSAT, the payload must be positioned on the nadir side of the spacecraft. Since it is sensitive to shocks, it is located as far as possible from the separation mechanism, which is situated at the bottom of the spacecraft. To avoid contamination of the payload by exhaust gases, the propulsion subsystem is placed at a large distance from the payload, naturally on the side opposite to the direction of motion [87, Table 14.6].

Another benefit of placing the propulsion subsystem at the bottom of the spacecraft is the reduction of the center of gravity (CG) position. As the propellant tanks are stacked vertically inside the central

CFRP cylinder, the propulsion assembly becomes vertically elongated, making its placement a major configuration driver. Since the S/C does not require strong control authority from the thrusters, they can be conveniently placed on a single panel facing $-x$, thereby simplifying hydraulic line routing. The offset of the thrusters with respect to the cylinder axis suggests that the spacecraft CG is shifted toward $-z$, opposite to the payload location. This indicates that the payload mass is relatively low compared to that of the other spacecraft components.

External sensors and antennas are mounted according to their field-of-view requirements. The relatively large external surface area of GOSAT provides ample mounting locations, allowing these components to be positioned with minimal impact on the internal spacecraft layout. Redundant units, such as X-band antennas and Sun sensors, are typically placed in close proximity to each other to simplify integration and ensure consistent coverage. Star trackers are oriented away from the Sun during nominal operations to prevent direct illumination and maintain measurement accuracy.

Solar arrays are mounted on the lateral sides of the secondary structure, providing a stiff attachment to the spacecraft bus. Since the dynamic behavior of deployed solar arrays can significantly influence the overall structural response of the spacecraft, their natural frequencies should be sufficiently high. This is achieved through a relatively low aspect-ratio design, which increases panel width while minimizing unsupported length within the constraints of the stowed configuration. The resulting increase in bending stiffness reduces array flexibility and limits dynamic coupling with the AOCS. Furthermore, the support struts position the solar arrays away from the spacecraft body, reducing the risk of shadowing and ensuring correct solar illumination throughout the mission.

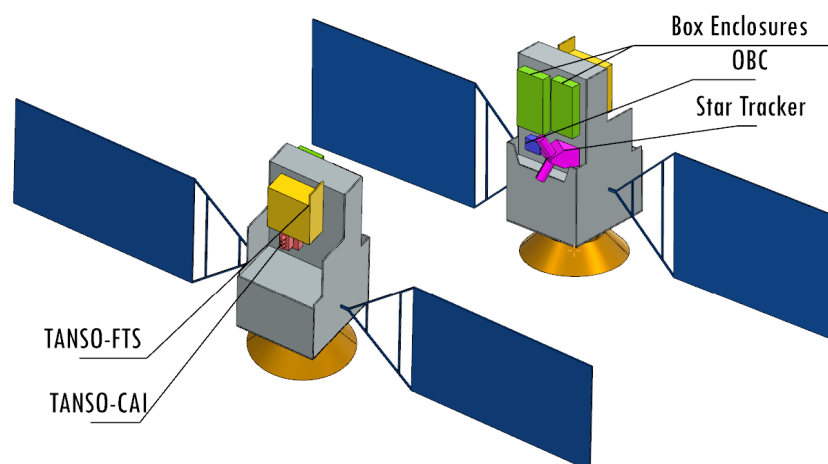


Figure 22: External component placement.

Most of the available internal volume is located above the primary structure and behind the payload section. To simplify routing, integration and testing, subsystems are likely grouped by functionality. Based on external observations, the following assumptions can be made:

- The two upper external boxes likely contain the batteries, other EPS components, and the TT&C subsystem. Their volumes are consistent with the required battery capacity and communication electronics, while the surrounding cabling suggests power-distribution and data-routing functions. External mounting also facilitates thermal control through radiator coupling. This centralized arrangement reduces the overall length of the electrical harness.
- A smaller centrally mounted box below them could contain the onboard computer, protected by a dedicated external cover.
- The reaction wheel assembly is likely accommodated within the main spacecraft body behind the payload. However, this location is not ideal, as wheel-induced vibrations can couple directly into the payload structure.

12.3 Launcher IF

12.3.1 Mechanical IF

The satellite is attached to the launch vehicle via the Payload Attach Fitting (PAF). Separation is achieved by firing pyrotechnics installed within the launcher interface. As confirmed by the launcher manual, the H-IIA vehicle utilizes a clamp band mechanism on the adapter; at the moment of spacecraft separation, the bolts connecting the band are severed by pyrotechnic cutters. Given this pyrotechnic separation requirement, the adapter model 1194M was assumed based on the limit loads and mass constraints provided in the launcher manual.

12.3.2 Launch loads

During the liftoff and in-flight phases, the GOSAT spacecraft is exposed to static and dynamic loads caused by the launch vehicle. The spacecraft structure is designed and tested to withstand these conditions:

- **Combined Loads:** The maximum loads on the primary structure occur at liftoff and at main engine cutoff (MECO). According to the 3-sigma limit levels, the maximum longitudinal acceleration reaches -39.2 m/s^2 (-4.0 G) immediately before MECO. During liftoff, the structure must withstand a combined lateral acceleration of $\pm 17.7 \text{ m/s}^2$ ($\pm 1.8 \text{ G}$).
- **Vibration Environment:** The spacecraft is exposed to low-frequency sinusoidal vibration and high-frequency random vibration. To avoid dynamic coupling modes in the low-frequency range between the launch vehicle and the spacecraft during ascent, the spacecraft's primary structure is designed with a structural stiffness ensuring fundamental frequencies of $\geq 10 \text{ Hz}$ in the lateral direction and $\geq 30 \text{ Hz}$ in the longitudinal direction.
- **CG Constraints:** The spacecraft's center of gravity (CG) position is strictly limited both in height from the separation plane and in radial offset from the thrust line of the launch vehicle [9]. This constraint is crucial, as the CG location directly affects the induced bending moments during high-acceleration phases and the angular rates at spacecraft separation. The CG's height limit from the separation plane is defined according to the structural strength requirements of the specific payload adapter. In particular, given the selected 1194M adapter model and the mass of the spacecraft, the allowable CG height limit is estimated at approximately 3.6 m [7, Figure A4.1-9]. Furthermore, the radial position of the CG must be maintained strictly within 25 mm of the launch vehicle's central longitudinal axis [7]. This extremely tight tolerance is mandatory to prevent excessive lateral loads and dynamic imbalances on the interface during the ascent.
- **Verification:** The spacecraft is required to withstand the maximum expected flight loads multiplied by minimum factors of safety. An environmental test report documents the adequacy of the spacecraft structure for these flight loads.

12.3.3 Fairing envelope

During the ascent phase, the satellite is encapsulated within a 4S-type fairing (4 meters in diameter, 12 meters in height) [7] [88]. This structure shields the spacecraft from aerodynamic drag, aerothermal heating, and severe acoustic vibrations from liftoff until atmospheric exit. The specific allowable envelope of the spacecraft inside the fairing is provided by the launcher manual [7, Figure 4.5-2].

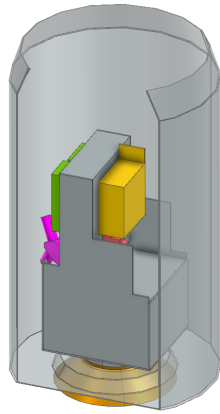


Figure 23: GOSAT
inside fairing

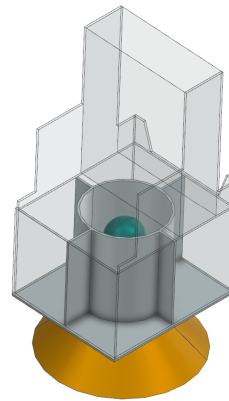


Figure 24: Semi-
transparent view

12.4 Mechanisms and Deployable Elements

GOSAT is equipped with numerous mechanisms, primarily located within the payload and solar array assemblies. During the critical deployment phase, six HDRMs in each wing are activated, allowing the solar arrays to deploy under the guidance of control wires. Once fully deployed, hinge-latch mechanisms secure the panels in their final configuration, with the control wires providing preload [89].

The S/C is equipped with a redundant set of panel rotation mechanisms acting about the y-axis, enabling robust operation in the event of a malfunction [90].

It is important to note that solar array wings are highly flexible structures and are therefore sensitive to thermally induced deformations and vibrations during orbital transitions between sunlight and eclipse. Studies on GOSAT have shown that sudden thermal gradients can induce quasi-static bending and low-frequency oscillations of the solar array paddles, which may affect spacecraft attitude stability [89]. As a result, the paddle rotation mechanism is required to have high structural stiffness to limit elastic deformations and reduce their impact on attitude stability under thermal and dynamic loading conditions.

12.5 Mass

An estimate of the structural subsystem mass was performed using a statistical approach. Although the reconstructed geometry includes approximately 36.67 m² of CFRP honeycomb structural elements with an average thickness of 30 mm, the model does not account for secondary structures, local reinforcements, brackets, fittings, solar array deployment mechanisms, launcher-interface hardware and other structural details. As a result, a direct mass computation would likely underestimate the total structural mass.

Therefore, a statistical correlation was adopted. For LEO spacecraft with propulsion, the structural subsystem typically represents approximately 27% of the spacecraft dry mass [9, Table A-6]. Assuming a GOSAT dry mass of 1635 kg according to Assignment #2, the structural subsystem mass is estimated to be approximately 441 kg. This value is considered representative of the complete structural subsystem, including both primary and secondary structural elements as well as associated mechanical hardware.

13 On Board Data Handling (OBDH) Architecture

On Board Data Handling (OBDH) is the central computing subsystem of the satellite, providing telemetry and payload data processing and storage prior to downlink, as well as command execution and subsystem health monitoring. Publicly available documentation for GOSAT does not identify a standalone OBDH system, instead the spacecraft block diagram show a split data handling architecture: a highly reliable chain for command and housekeeping, and a separate high rate branch for the TANSO payload mission data [16].

13.1 Components

The telemetry branch follows the data flow from the redundant Remote Interface Module (RIM)-A/B, which collect data from each subsystem, to the Housekeeping Data Recorder (HKDR), which stores the data until a communication link is available, and finally to the redundant Command and Data Management System (CDMS)-A/B, which act as the interface with the S-band antennas. The opposite sequence applies in the case of command uplink.

The mission data branch instead relies on the Mission Data Handling Subsystem (MDHS), which receives the measurements from the PLs and records them on the semiconductor memory dedicated to the scientific data through the Mission Data Processor (MDP), before they are played back through the X-band [16].

13.2 On-Board data memory

The On-Board data memory is mainly dedicated to storing the scientific data generated by the TANSO system, with a smaller fraction reserved for housekeeping data. A total capacity of 48 GB is allocated [16], ensuring sufficient storage capability even in the event of a missed connection with an X-band ground station. As for the telemetry, the maximum no visibility period must not exceed 6 hours in the case of station malfunction [51], which requires the OBDH system to be capable of storing this data volume. However, since housekeeping data generation is negligible compared to payload data volume, as later shown in section 14, this requirement is satisfied by meeting the X-band missed connection data storage capability.

13.3 Bus Architecture

The GOSAT onboard data architecture is considered a hybrid centralized bus architecture, where different interfaces are selected according to the data volume being transferred. Three main interfaces are used:

MIL-STD-1553B architecture is used for command and telemetry bus due to its proven reliability and deterministic data delivery [27]. It can easily handle data rates in the order of kbps and it is a dual-redundant architecture that satisfies the S/C requirement to avoid single point failure.

RS-422 is selected for point-to-point moderate data rate payload like TEDA, CAMs, and the electronics for the AOCS [27]. This avoids unnecessarily loading the MIL-STD-1553B bus with local unit traffic, and provides a simpler architecture with respect to the SpaceWire.

SpaceWire is assumed for the TANSO payload, since no specific protocol is highlighted in [27], due to the high scientific data rate of approximately 7 Mbps. This selection is also justified by the LRO OBDH architecture, a mission launched in the same period as GOSAT, which employs the same three protocol strategies to handle different data rates [91].

13.4 On-Board Computer (OBC)

In GOSAT's architecture, the On-Board Computer (OBC) serves primarily as the bus controller on the 1553 network, issuing commands to the S/C's subsystems and collecting housekeeping telemetry, while it communicates directly with the RS-422 linked instruments and secondary payload units. Instead for TANSO payload data, SpaceWire relies largely on the MDP, with the OBC being responsible for initiating data flows and managing onboard data storage rather than actively forwarding traffic.

The OBC hardware is typically based on a radiation-hardened 64-bit MPU such as the hr5000, a 320 MIPS class processor representative of JAXA's mid-2000s space computing generation. This processor executes a JAXA-developed Real-Time Operating System (RTOS) derived from the TOPPERS/HRP kernel, ensuring deterministic scheduling and fault containment [92, 93]. Together, the MPU and RTOS provide the real-time responsiveness and fail-safe operation required for autonomous spacecraft control and long-duration mission reliability.

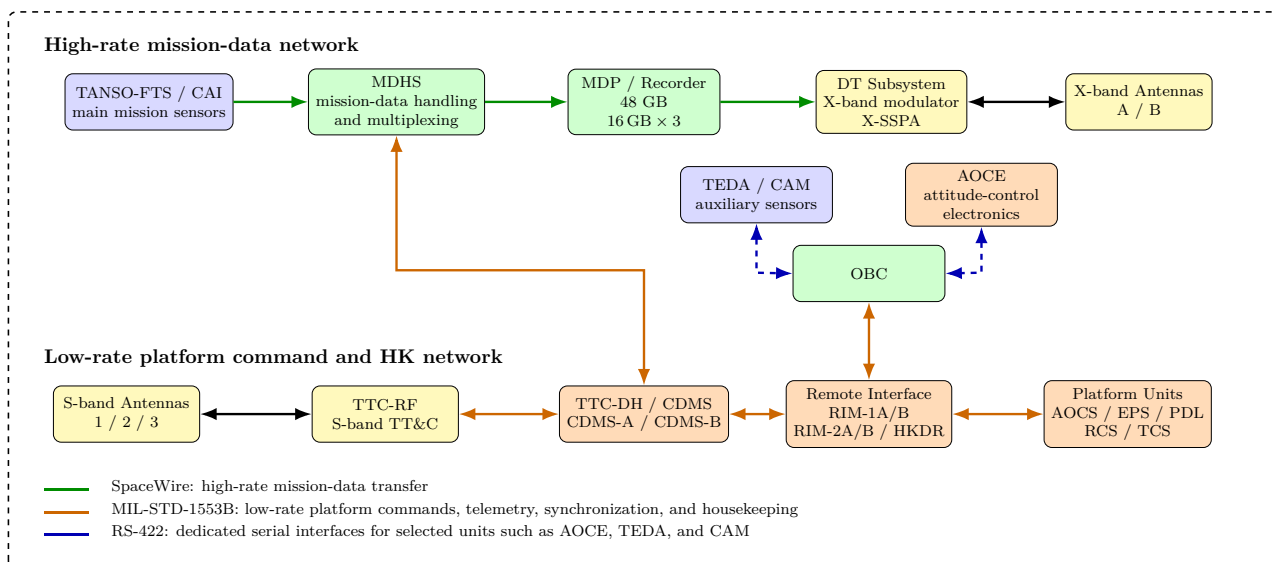


Figure 25: Simplified GOSAT onboard bus architecture

14 Data storage

To estimate the data volume produced by the S/C during nominal operation, the following assumptions for each subsystem are adopted, based on a reasonable selection of variable data types, since no detailed breakdown is available in the literature.

PS

In the PS, pressure and temperature of both tanks have to be measured, to maintain hydrazine in safe conditions as well as to estimate the percentage of remaining propellant. Considering a resolution of 0.001 bar and 0.01 K respectively, the `uint16` data type allows to represent quantities up to 65.535 bar and 655.35 K, which cover completely the ranges expected in this section of the S/C. Additionally boolean variables are used to store the state of valves and catalytic bed heaters.

TMTC

The current and voltage supplied to the amplifiers are monitored to compute the power delivered to the antennas, while two boolean variables indicate when uplink or downlink connections are estab-

lished.

Attitude and Orbit Control System (AOCS)

Apart from the payload, the AOCS is the subsystem generating the highest data volume due to its higher update rate compared to housekeeping data. In particular since the update rate of the STT is 10 Hz, the AOCS loop has been assumed to work at the same frequency. To ensure high precision all vectorial quantities are stored as `float32`.

EPS

The EPS data volume is mainly composed of housekeeping data, monitoring the voltage and current of the solar arrays, batteries and bus. As the main bus operates at 50 V DC, a `uint16` is sufficient to represent both voltage and current with a resolution of 0.001 (V or A) up to a maximum value of 65.515 (V or A). Moreover the orientation of the solar arrays and its control input for the paddle motors is stored and sampled at 1 Hz, as the rest of the housekeeping information, due to its relatively slow variation.

Payload (PL)

The payload is the main contributor to the data generated on board, with FTS and CAI having 5.9 Mbps and 1.12 Mbps data rates respectively. In order to compute the data rate in the sunlight phase both PLs are considered active whereas in the night phase CAI observations are suspended and only one of the four bands of FTS (TIR) is active, leading to a reduced data volume.

TCS

Considering that all the temperature sensors have been allocated in each subsystem data volume, only the current used by active heating/cooling systems and their state is considered in the TCS data rate. As mentioned in the TCS section, a total of 250 temperature sensors are considered, of which only 72 are associated with this subsystem as the remaining are allocated in the others.

On Board Data Handling (OBDH)

For the OBDH subsystem, in addition to the health monitoring parameters, the watchdog timer as well as the on board time have to be stored. Both are stored as integers in order to avoid rounding errors and always ensure their correct behaviour.

Once the total data rate is computed, the estimation of the data volume produced during one orbit must account for the fact that different observation occur during daytime and nighttime.

Considering that during the sunlight phase ($T_s = 3802.32$ s) both instruments are active, whereas during eclipse ($T_e = 2085.12$ s) the CAI cannot acquire measurements and only the TIR channel of the FTS remains operational, corresponding to one out of four bands, a reduced FTS data rate equal to 1/4 of the nominal value is assumed during eclipse, leading to a total data volume of 3.73 GB per orbit, as shown in Table 21.

Adding a 400% margin, the computed final data production per orbit is 14.96 GB. The values obtained are consistent with the 48 GB memory allocated to scientific data, as this capacity allows the storage of multiple orbital observations until an X-band downlink station becomes available.

Subsystem	Parameter	n.	Data Type	Size [bit]	Frequency [Hz]	Data Rate [bps]
PS	Tank Pressure	2	uint16	16	1	32
	Tank Temperature	2	uint16	16	1	32
	Valves on/off	18	boolean	8	Event	–
	Cat. bed on/off	12	boolean	8	Event	–
	Cat. bed temp.	12	uint16	16	1	192
TMTC	S-band amp. current	1	uint16	16	1	16
	S-band amp. voltage	1	uint16	16	1	16
	X-band amp. current	1	uint16	16	1	16
	X-band amp. voltage	1	uint16	16	1	16
	Lock flag (S/X band)	2	boolean	8	Event	–
	Amp. and ant. temp.	10	uint16	16	1	160
AOCS	Estimated quaternions	1 (4x1)	float32	128	10	1280
	Angular velocities	1 (3x1)	float32	96	10	960
	Sun direction	1 (3x1)	float32	96	10	960
	Nadir direction	1 (3x1)	float32	96	10	960
	AOCS mode	1	uint8	8	Event	–
	STR output	1 (4x1)	float32	128	10	1280
	HOR output	1 (3x1)	float32	96	10	960
	FSS output	3 (3x1)	float32	96	10	960
	IMU output	1 (6x1)	float32	192	10	1920
	GPSR output	1 (3x1)	float32	96	10	960
	RW velocity	4	float32	32	10	1280
	RW sat. flag	4	boolean	8	1	32
	RW command	4	float32	32	10	1280
	MTQ current	2	uint16	32	10	640
	MTQ command	2	uint16	32	10	640
	Temperature	25	uint16	16	1	400
EPS	Battery current	8	uint16	16	1	128
	Battery voltage	8	uint16	16	1	128
	Battery temperature	8	uint16	16	1	128
	Bus voltage	1	uint16	16	1	16
	Bus current	1	uint16	16	1	16
	Active bus selection	1	boolean	8	Event	–
	Array deploy status	2	boolean	8	Event	–
	Array paddle angle	2	float32	32	1	64
	Paddle control input	2	float32	32	1	64
	Array current	2	uint16	16	1	32
	Array voltage	2	uint16	16	1	32
	Array temperature	30	uint16	16	1	480

Subsystem	Parameter	n.	Data Type	Size [bit]	Frequency [Hz]	Data Rate [bps]
PL	TANSO-FTS	1	–	–	–	5900000
	TANSO-CAI	1	–	–	–	1120000
	Payload voltage	2	uint16	16	1	32
	FTS obs. mode	1	uint8	8	Event	–
	CAI obs. mode	1	uint8	8	Event	–
	Payload temperature	100	uint16	16	1	1600
TCS	Heater on/off	5	boolean	8	Event	–
	Active coolers on/off	5	boolean	8	Event	–
	Heater current	5	uint16	16	1	80
	Coolers current	5	uint16	16	1	80
	Other temp. sensors	72	uint16	16	1	1152
OBDH	Watchdog counter	1	uint16	16	10	160
	OBC current	1	uint16	16	1	16
	OBC voltage	1	uint16	16	1	16
	OBC temperature	3	uint16	16	1	48
	OnBoard Time	1	uint32	32	10	320
	Boot cause	1	boolean	8	1	8
Total datarate [Mbps]		-	-	-	-	7.04
Data per orbit [GByte]		-	-	-	-	3.74
+ 400% Margin [GByte]		-	-	-	-	14.96

Table 21: Data rates per subsystem

Subsystem	Component	Margin	Mass w/o margin [kg]	Mass w/ margin [kg]	Source
PS	Propellant Tanks	5%	6.694	7.028	S
	20 N Thrusters [22]	5%	2.88	3.024	D
	1 N Thrusters [22]	5%	3.04	3.192	D
	Pressurant	20%	1.213	1.456	S
TMTC	X-band ant. [37]	5%	0.65	0.683	D
	X-band ampl. [39]	5%	4.1	4.305	D
	S-band ant. [47, 48]	5%	0.435	0.457	D
	S-band ampl. [40]	5%	2	2.100	D
AOCS	STT [54]	5%	0.74	0.777	D
	ES [55]	5%	3	3.15	D
	GPSR [61]	5%	0.062	0.065	D
	FSS [56]	5%	2.25	2.363	D
	IRU [57]	5%	6.6	6.93	D
	MTQ [59]	5%	7.6	7.98	D
	RW [58]	5%	34.4	36.12	D
EPS	Battery	10%	241.92	266.11	S
	Solar panels	10%	107.58	118.34	S
PL	TANSO-FTS [64]	20%	329	394.8	D
	TANSO-CAI [65]	20%	42	50.4	D
	CAMs [94]	5%	41.6	43.68	D
TCS	[-]	[-]	32.7	32.7	E
OBDH	[-]	[-]	87.5	87.5	E
STR	[-]	[-]	441	441	E
Total dry mass w/ component margin: 1514.16 kg					
Total dry mass w/ 20% system level margin: 1816.99 kg					
Propellant mass: 115 kg					
Total wet mass: 1931.99 kg					
Adapter mass [7]: 100 kg					
Total launch mass w/ adapter: 2031.99 kg					
Launchable mass [7]: 4200 kg					

Table 22: Computed Mass Budget. D = datasheet, S = sizing, E = SMAD estimate [9].

Subsystem	Component	Mass [kg]
PS	Propellant Tanks	[-]
	20 N Thrusters	[-]
	1 N Thrusters	[-]
	Pressurant	[-]
TMTC	X-band ant.	[-]
	X-band ampl. [39]	4.1
	S-band ant.	[-]
	S-band ampl.	[-]
AOCS	STT	[-]
	ES	[-]
	GPSR	[-]
	FSS	[-]
	IRU	[-]
	MTQ	[-]
	RW	[-]
EPS	Battery	[-]
	Solar panels	[-]
PL	TANSO-FTS [64]	329
	TANSO-CAI [65]	42
	CAMs [94]	41.6
TCS	[-]	[-]
OBDH	[-]	[-]
STR	[-]	[-]
Total dry mass: 1635 kg		
Propellant mass: 115 kg		
Total wet mass: 1750 kg		
Adapter mass [7]: 100 kg		
Total launch mass w/ adapter: 1850 kg		
Launchable mass [7]: 4200 kg		

Table 23: Literature Mass Budget.

15 Mass Budget Considerations

A comparison between the total mass values in Table 22 and Table 23 reveals a close correspondence. This alignment is expected, as approximately one-third of the preliminary sizing mass budget is derived from the empirical estimation standards found in [9], starting from the literature dry mass itself.

To account for the differing maturity levels of the subsystems, specific Design Maturity Mass Margins were applied at the equipment level:

- 5% for off-the-shelf products without modifications (Product Categories A and B, e.g., AOCS and TMTC components);
- 10% for off-the-shelf products with modifications (Product Category C, e.g., the EPS battery and solar panels);
- 20% for newly designed and developed products (Product Category D, such as the payload instruments).

Additionally, a 20% margin was applied to the mass of all pressurants within the pressure-fed propulsion system. To conclude the sizing envelope, a 20% system-level mass margin was applied to the nominal dry mass at launch to determine the total dry mass at launch of the spacecraft.

The actual wet mass of the S/C (1750 kg from literature) falls comfortably between the computed dry mass (1514.16 kg) and the total wet mass including system-level margins (1931.99 kg). This placement validates the conservative margin strategy applied. Because these margins were established during the preliminary design phase, they are intentionally generous. This high margin threshold is necessary because the preliminary sizing does not explicitly model components that are difficult to quantify at this stage, such as harness cabling, routing connectors, and small electronics or feed lines. Consequently, the applied equipment and system-level margins successfully absorb these unmodeled elements, keeping the actual spacecraft mass safely below the maximum design envelope.

While these overall-level values align well, a more detailed, subsystem-level validation against the actual flight hardware was hindered by the limited public availability of technical documentation for GOSAT.

Nonetheless, on a launch integration level, the overall spacecraft mass (approximately 2032 kg with margins and adapter) remains significantly below the maximum launchable capacity of the H-IIA rocket. This substantial performance margin allowed the launch operators to optimize capacity by hosting several secondary piggyback payloads accounting for approximately an additional 238 kg of launch mass [1].

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